

Question:- Biosafety (2)

Answer:-

- prevention of large-scale loss of biological integrity
- prevention mechanisms
- conduction of regular reviews - biosafety in laboratory settings
- Strict guidelines to follow

Question:- Biosafety rules

Answer:-

- biosafety guidelines ministry of env
- commercial release
- deliberate release
- experimental release

Question: What are surveillance elements?

Answer: heat detector, smoke detector, lighting, alarms, cameras

Question:- Surveillance and notification system:

- These are the **heat detectors** Heat detectors can give us alarms there is high temperatures for samples as well as equipment then
- **Smoke detectors** detect the smoke it protect the building from fire.
- **Light** is proper in lab especially beech work then emergency alarms should be there
- **Cameras** to control the access of authorized people to the lab.

Question: Genetic modification:

- any alteration of genetic material
- capable of producing new substances

- performing new functions

Question:- Name the things for protection in first aid kit (2)

Answer:- It includes Mouth piece for mouth to mouth resuscitation then gloves are Important in order to protect hands from carrying infectious material. Clean up kit for blood spills or any container spillage, there must be a disinfectant.

- first aid box
- protective clothing
- safety equipment
- eye irrigation

Question:- Name lab safety things 2

Answer:- Lab safety

Safety glasses, closed-toed shoes, no food/drink, long hair must be tied, lab coat, open flames-unattended, flammable liquids, skin contact-rinse off, proper waste disposal, liquid waste-labeled containers, equipment must be cleaned and placed back, inform-chemical spill/thermometer breakage, be careful-handling hot glassware and apparatus, avoid taking excess amount of chemicals, cell phones/ head phones are not allowed.

Question:- Cybernetics (2)

Answer:- The study of interactions between man, machine and animals is called cybernetics. It is a Greek word “governance”. It is exploring regulatory system, their structures and functions.

Question:- Risk assessment made by GMO 2

Answer:-

- Ist the Risk assessments of field trials to exporting country then you have to give approval to the applicant. We have to follow rules of national plant quarantine regulations or international plant protection convention.
- In the form of genetically modified organism such as plant, it is resistant to herbs it can be resistant to insects, then you want to import such products that is not available in country or genetically modified substances or cells you want to import from other countries.
- For the import export of genetically modified organisms or modified living products we have to follow certain requirements in accordance with article 18 of Cartagena protocol as set forth in biosafety guidelines.

Question: TAC function

Answer:- Technical Advisory Committee(TAC) they not only deals with biosafety rules also deals with Biosecurity rules depending upon field which GMOs are used.

Question: Radiation effect minimized

Answer:-

- minimize the time of exposure to radiation
- maximize the distance from the radiation source
- shielding the radiation source
- substituting with non-radiometric methods

Question:- Biosafety level

Answer:- **Definition:**

Level of containment precautions, isolate dangerous biological agents, enclosed laboratory facility, containment level (BSL-1 to BSL-4)

Question: Facilities apparatus in BSL1

Answer:- Facilities Apparatus

Doors for access controls, sink for hand washing, bench tops-resistant, chairs-easy to disinfect, lab windows fitted with screens.

Question: Containment zone elements

Answer:- **Containment zone:**

Only be a chemical fume hood, isolation of microorganisms, building systems, sealed rooms, sealed containers and personnel suits.

Question:- Write some activities of bio safety officer.

Answer:- **Activities:**

Biosafety consultation, biosecurity, consultation, technical compliance, consultation.

Question:- Physical security?

Answer:- The physical events that are of terrorism and disasters being used the component of biosecurity that is called physical security.

- protection of people/ hardware/programs

- data / networks
- physical events-terrorism/disasters

Question:- Physical Security elements? 3

Answer: It not only deals with protection of people but Also deals with hardware network programmers as well as protection of data .Now physical elements are of three types these are;

- obstacles
- notification and
- recovery stage.

Question:- Functions of physical security?

Answer:-

- First component of biosecurity it gives not only protection to people working in the lab .It also protect hardware or the programmers with which organization the lab work dealing with then data and network.
- Physical events and physical circumstances that can occurs it can occurs in the form of terrorism then natural disasters in the form of fires or explosive chemical physical security not only deal with protection of people lab workers. In order to deal with the physical events that is of terrorism and disasters be used the component of biosecurity that is physical security.

Question: Biosecurity? 3

Answer:- A set of preventive measures designed to reduce the risk of transmission of infectious agent's. Security against the inappropriate use of potentially dangerous biological agents.

Types:

- laboratory biosecurity programs
- animal biosecurity
- Bioweapons

Question:- Elements of act?

Answer:- **Act:**

- recruitment
- transport
- transferring
- harboring
- receipt of person

Question: Biosecurity plan (3)

Answer: Biosecurity plan:

- **written plan-prevent the introduction and spread of disease to farm**
- **daily operation procedures**
- **disinfecting procedures-part of the plan**

Written plan prevents introduction and spread of disease. It is written documentation or written plan that prevents introduction of pathogens into nation that affect any population or sub population even it can effect organizations. It not only deal production of pathogen in to the environment or organizations also prevents spread of diseases. We have to look daily operations procedures also deals with disinfection method important part of biosecurity plane. Biosecurity plane start with disinfection.

Question:- Explain different type of fire extinshuer 3

Answer:- Types of fire extinguishers:

- Water and foam –class A – separate oxygen
- Carbon dioxide- class B and C – separate oxygen and heat
- Dry chemical – class A, B, C- interrupt chemical reaction
- Wet chemical-class K – remove heat
- Clean agents-class A, B and C (halogens) interrupt chemical reaction
- Water mist- class A, remove heat

Question: Equipment used in BSL3, write 5 points (5)

Answer:- Equipment:

Vacuum lines must be protected with HEPA filters

Now here special equipments are required for example apart from personal equipment galss eye feelany shoe cover we also require special vacuum line must be protected with HEPA filters.

2. Ducted air ventilation system

WE need Ducted air ventilation system. In biosafety cabinet HEPA filters used same in ducted air ventilation system. In biosafety cabinet actually a chamber that has three filters 1 hepa filter intake air and

3. Hepa filter for decontamination

Hepa filter for exit air in order to dispersed away the contaminated air from occupied area.

4. HEPA filter exhaust air BSL-3 facility design

BSL 3 must be fully documented facility design that include decontamination of larger pieces of equipments any sort of operation any parameter regarding procedure must be documented prior to performing the procedure.

5. Operational/parameter/procedures must be documented

Question: - Measures required for TB lab?

Answer: - Biosafety measures for TB lab:

There should be double set of door. Doors one for entry and other for exit.

Children and unauthorized peoples are not allowed. Self-closing system of doors.

Biohazard symbol should displayed on the door.

Worker working in lab must follow personal protective equipment's gloves protective eye wears shoes cover and hair covers.

After performing experiment wash hands. Clean bench top with 70% alcohol.

If using BSC then check HEPA filters check maintains daily.

Decontaminate and waste disposal.

Question:- Biohazard is classified into four elements name them write their categories, name their substance

Classification:

- Category A, UN 2814-infectious substances affecting humans
- Category A, UN 2900-infectious substances affecting animals
- Category B, UN 3373- biological substances transported.
- regulated medical waste, UN 3291- waste or reusable material
- derived from medical treatment/research

Question:- How immunization can be provided to fight against pathogens. 5

Answer:- Immunization of staff

Discuss with workers

This is for personal safety and safety of worker. Immunization is an example of active immunity which is very necessary if you want to do work in the laboratories. If you appoint any person in lab then you have to check the vaccination card whether they fulfill all the vaccinations included EPI programmer or not also **discuss with workers.**

2) Vaccines

Vaccines are available for hepatitis B that should given to the workers very important viruses in Pakistan Mutation not cause by immunological genetic factor but also cause by environmental factors microorganism is important part of environmental factors they bring about the mutation and cervical cancer is one of major cancer 80% female suffer from cervical cancer for that HPV vaccination available vaccination available for human papilloma virus it can cause cervical cancer in female we can get vaccination from age of 19 year and onwards.

3) Therapeutic drugs-after exposure

If exposure in lab then **therapeutic drugs** should be available in lab in order to treat workers.

Question:- Functions of technical adversary committee? 5

Answer: Function of technical Advisory Committee:

- **Examine applications and recommend to NBC:** Examine applications researchers about the project and recommend it to national biosafety committee for their approval they review control safety measures as well.
- **Review and control of safety measures:** Look at safety regarding humans and environment as well.
- **Review research methodologies:** Review research methodology as well.
- **Monitor release of GMOs/products into environment:** They also monitor commercial release of GMO and products into the environment.
- **Provide information to NBC about approved projects:** Provide information regarding national biosafety committee about approved projects up to which level project have been completed funds utilized.

- **Supervise the implementation of terms and conditions:** Also supervise implementations of terms and conditions

Question: First aid equipment's? 5

Answer:-

First aid must contain aid box, protective clothing (contain gloves, eye shield, face mask) safety equipment and eye irrigation. Safety equipment can be in the form of rope for fire hazards, mouth to mouth pipe for resuscitation then eye irrigations system must be properly placed in the first aid box.

First aid box:

- made up of material-keep the content dust and damp free
- white cross with green box
- prominent position
- easily recognizable

Box contains:

first aid manual, bandages, sterile dressings, safety pins

Protective equipment:

mouth piece for mouth-to-mouth resuscitation, gloves-protections, clean up kit for blood spills.

Question:- What are functions institutional biosafety committee 5

Answer:-

The Institutional Biosafety Committee (IBC) is a committee created under mandate from the National Institutes of Health (NIH).

- The IBC is responsible for enforcing policies and guidelines related to university-related use of all potentially hazardous biological agents including but not limited to infectious agents, human and non-human primate materials.
- The Committee ensures that research involving these agents is conducted in a manner that complies with all applicable regulations, policies and standard and does not endanger the researcher, laboratory worker, human research subjects, the public or the environment.
- The IBC is responsible for reviewing projects that involve, but are not limited to, recombinant DNA, RNAi, pathogens, human materials and other potentially infectious material, as well as transgenic animals.

- The IBC provides recommendations to the intramural community in matters pertaining to the control of biohazards associated with the use of microbiological agents and their vectors.
- It also represents the interests of the surrounding community with respect to public health and protection of the environment.

Question:- WHO classification of pathogens. 5

Answer:-

- WHO risk group 1: microbes unlikely to cause disease
- WHO risk group 2: microbes causing diseases-unlikely to be serious
- WHO risk group 3: pathogens causing serious disease
- WHO risk group 4: pathogens causing serious disease ---- transmission-----no effective treatment or preventive measure

Question:- Discuss transport security element.

Answer:- Transport security Elements

It is one of the components of biosecurity. It consist of two elements one is internal transport other is external transport.

Internal transport:

It is transport of movement of biological materials with in the country. Movement of biological material from to restricted area but remain within the country. If we are moving biological material within the institution we will refer reliable lab works but if we moving biological material from one city to another we will divide the area into three groups

- 1st is the Shipping Area: It is the area where disposing biological materials for transport here use your id card special documents regarding the composition of agents or pathogens.
- 2nd is the Receiving Area: Here the receiving name should be mentioned; he/she will bring ID card, and has to show answer of the code question.
- And 3rd one is disposal area: That is incinerator used for pathogens and toxins.

External transport:

It is the transport of biological material to another country these are two elements of transport biosecurity. Movement of biological material from one facility to another. So this involves commercial carriers. More than 30 commercial carriers are available and the most commonly used is IATA and WHO. Here we are able to move frozen samples material from one place to another. External transport is expensive as compared to internal. In external transport we divide material into two groups first group is category A second one is category B infectious material or pathogens or their toxins in the form of culture, isolated DNA, isolate RNA, or in the form of isolated nucleic acids. It is included in category B. Now such infectious material required triple packaging system; triple packaging names indicates consists of three layers; primary, secondary and tertiary layers.

- Primary layer consist of specimen that protects the specimen any sort of leakage.
- Secondary layer protect the primary layer and **water proof and**
- Third Tertiary layers protect the secondary layer.

Question:- Future aspect of this biosecurity act 2015

Answer:- Biosecurity act support biosecurity system in any age. Regardless of the advances in technology in scientific field like biotechnology. **Scientific advances and advises** help to make right decision.

Objectives: Was to create a modern and responsive legislation framework. Improves the underpinning process so create the robust biosecurity system that benefits everyone. Now this robust system prevent entry and establishment of invasive and exotic pests which are lethal because these species not only harmful for the environment but also related to the safety of humans also depends upon economy.

Question: Write note on special practice in BSL2. 10

Answer:- BSL2 Special Practices

1. Meet specific entry and exit requirements

Door must be kept close that meet specific entry and exit requirements. The people were entering in the lab must be aware infectious agent.

2. Workers must be immunized

The workers which are working in the lab must be immunized against infectious diseases such as hepatitis B.

3. Biosafety manuals must be available

Biosafety manuals must be accessible and available it is duty of lab manager.

4. Proper collection, handling, processing, storage or transport

While collecting the sample we must take care there should be proper collection of specimen in a durable leak proof container. Then proper handling and processing of specimen storage must be done at specific temperature some specimen freeze at 40 and some -40degree centigrade. If you want to transfer specimen from collection point to the diagnostic lab you should have to maintain chemical composition of specimen.

5. Lab equipments must be decontaminated

Lab equipment must be decontaminated after performing procedure. If any sort of incident occurs in the lab for example any biological spill occurs it must be informed to the biosafety officer. He can improve biosafety levels.

6. Incidents must be informed/eye, face, hand protection

Eyewear face cover must be wear whenever you want to perform microbiological practices. If you wear a lab coat you have to take it off if you are going cafeteria library or administrative office.

7. Animals and plants should not be permitted in the lab

Animals and plants not allowed in the lab even small children because they have weak immunity

8. Aerosols-physical containment equipment

If there are chances of formation of aerosols while shaking grinding centrifuging pipetting even sonicating opening of container having infectious material there are chances of aerosol generation so in order to avoid this all standard microbiological practices performed in a biosafety cabinet. Biosafety cabinet should be satisfy should be used according to manufacturer procedure and precautionary measures.

Question:- BSL3?

Answer:- BSL-3:

- **Applicable to diagnostic / clinical/ research/ production/ teaching facilities**
- **Potentially lethal disease through the inhalation route**
- **All procedures must be performed in a biosafety cabinet**

It is applicable to diagnostic lab clinical research areas even production and teaching facility's, BSL3 include those agents that cause fetal disease and potentially hazardous disease in the human being because they appear in the environment in the form of aerosols and human being can inhale them as result enter in to the body through inhalation rout. Microbiological practices should be performed in the biosafety cabinet.

Equipments:

- Vacuum lines must be protected with HEPA filters
- Ducted air ventilation system
- HEPA filter exhaust air BSL-3 facility design
- Operational/parameter/procedures must be documented

Now here special equipment's are required for example apart from personal equipment glasses eye shield and shoe cover we also require special vacuum line must be protected with HEPA filters. WE need Ducted air ventilation system. In biosafety cabinet HEPA filters used same in ducted air ventilation system. Biosafety cabinet actually a chamber that has three filters 1 HEPA filter intake air and 2 HEPA filter for exit air in order to disperse away the contaminated air from occupied area. BSL 3 must be fully documented facility design that include decontamination of larger pieces of equipment any sort of operation any parameter regarding procedure must be documented prior to performing the procedure.

Question:- BSL-4:**Answer:- BSL-4**

- Dangerous exotic agents
- Aerosol transmitted lab infections
- No vaccine/treatment
- Unknown risk of transmission

This is the highest containment level it includes those agents that are dangerous exotic because they have ability to appear in the environment or contaminate environment in the form of aerosol. Aerosols are easily inhaled by the workers who are working in the lab and can be easily infected by the infectious agents. These are dangerous because cause lethal disease in humans no special vaccines are available no special treatment no special precautionary measures can be taken. Here we have to take special precautionary measures for workers who are working in lab in order to prevent the formation of aerosol as much as possible now there is unknown risk for transmission.

We divided BSL-4 in two types

A cabinet laboratory: Manipulation of agent in BSC; be can manipulate culture infectious material in a biosafety cabinet. Now biosafety cabinet that is satisfied and used according to the manufacturer detail they have double doors material that contain infectious agent first passes through autoclave for sterilization then disposed of biosafety cabinet. Now in autoclave sterilization occurs 21 degree centigrade for 15 to 20 minutes pressure 15 pounds. Biosafety cabinet must be designed if infectious material not sterilized in autoclave it must be removed carefully from biosafety cabinet. Never put sharp things in BSC. It is better to avoid such material. Now apart from HEPA filters which are used in filtration that intake air and excrete air gas and vapor decontamination unit in BSC. Now the worker who are using biosafety cabinet must wear glass eye shields then shoes cover once they enter into the lab they have followed all those personal equipments which are required for the lab BSL4. It is responsibility of lab manager to look at repair and maintenance of

BSC. 2ND TYPE: A suit laboratory: Where personnel must wear positive pressure air protective suit. BSC used for all microbiological procedures and decontamination odd gloves and apparatuses during lab operations.

Question:- Risk management?

Answer:-

- **Article 15 Annex 3 of Cartagena protocol set forth in national biosafety guidelines.**
- **NBC will ensure**
- **Activate biosafety guidelines:** National Biosafety committee will monitor all activities that are performed by researchers whether they are according to biosafety guidelines or not.
- **Licenses:** NB committee also check whether the before starting the experiment did they got the license, or before dealing with the GMOs did they get the license from the federal agency or not.

Risk assessment:

- Auditing of risk assessment
- Evaluation of risk management measures

- Field trials

Auditing of risk assessment in Pakistan we followed the risk assessment scheme in which we compare the variables second method is scoring system in which we check out that whether activities are at which risk level according to data we access the risk but management is very important performs by three ways the first one is **auditing second one evaluation of risk management an third one is field trials which is carried out case by case studies in a scientifically sound manner.**

Question:- What are the activity of human which change the fresh water into sewage? (5)

Answer:-

Human activities change clean water into waste water. These activities are as follows

- Domestic
- Agriculture
- Commercial
- Industrial
- storm water
- runoff water

It can be domestic activity; it can be Agricultural, commercial, industrial, storm water, or it can be due the runoff water. In sewage water pass through a simple pipe but due to heavy rain sewerage can get overflow and water comes on the ground this can contaminates different areas and can be threatful regarding health of human being.

Question:- Applications of biosafety rule? 5

Application:

- manufacture, import and storage of microorganisms
- gene technological products for research
- field trial of GMOs
- import, export, sale and purchase of GMOs

Question: What are the precautions made in BSL2 as compared to BSL1 (10)

The major difference in BSL1 and BSL 2 workers which are working in BSL 2 in lab must have training for handling infectious material.

Special practices and equipment also required.

We restrict movement of people in the lab having BSL-2.

Then special bio containment required for this lab especially biosafety cabinet.

Question:- Bioweapons? And their threats and how they migrate 10

Answer:- **Effort To Mitigate Biological Threat**

Bioweapons:-

Whenever we use word biological threat term comes in our mind bioweapon or bioterrorism. Bio mean life and weapon means infectious agents that are used weapon against human being known as bioweapons. Or

Any infectious agent that bacteria viruses used intentionally to cause harm to human being known as bioweapon other term that used for bioweapons is bioterrorism these both are synonyms.

Now at national and international plane or programs developed against the bio war defense planning of effective bio war defense very difficult task whether national or international level. It affects the nations and bring fear among public also effect the scientific community.

Defense against Bioweapons:

- International cooperation
- Transfer of technology
- Support national actions

International cooperation we have to cooperate with international agencies to control bioweapons. Also **transfer of technology** that help to create bioweapons, we have to **Support national actions**. These three things are important that help to create defense against bioweapon.

BTWC:

- BTWC-1972
- institutionally weak

- implementation is ineffective without investigation

Biological threats:

Now let's talk about threats that can act as bioweapons.

Terrorist, non-state actors

So these biological threats are number one is Terrorists; they can be non state actors, either that can be misuse of technology, if we are not using good microbiological practices.

2. Misuse of technologies:

Then the misuse of technology can release the infectious agents into the environment.

3. Theft from lab

Infectious agents Bacteria or viruses that are acting as bioweapons they can be theft from lab.

4. Religious extremists

In the same way National and International religious extremists they can act as biological threat.

Locally hired agents

Sometimes for terrorism international agencies are involved but because of Un employment, poverty they locally hired agents so frustration among people increases due to poverty and unemployment in developing countries, usually such people are used to spread the bioweapons.

5. Frustrated cult

Potential threats:

naturally present in the environment, no major infrastructure, no manpower, easier and faster, cost-effective than poor's man atomic bomb, cover large area, difficult to diagnose and treat, high mortality and mobility, possibly contagious-small pox, plaque, viral hemorrhagic fever, create panic, weapon of mass disruption, detection devices, equipment for surveillance-expensive, not present in many countries

Efforts to mitigate:

BTWC Act

designation of focal point

central implementation authority

biosafety/biosecurity

code of conduct /awareness

According to BTWC ACT which was established in 1972, there must be designation of focal point, there must be central implementation authority dealing with biosafety and biosecurity guidelines. Then they talk about the code of conduct for biologists and for the researchers. The most important thing is awareness regarding biosafety, biosecurity and bioweapons among the public.

Question: What are malicious risk groups, its types explain it? (10)

Answer: **Malicious risk groups**

Non-pathogenic

These are inherent hazardous agents or pathogens that cause no or insignificant consequences. These are non-pathogenic organisms that are not harmful for human beings.

2. Low malicious use risk (LMUR)

Then the 2nd includes Low Malicious Use Risk LMUR IS SO CALLED low because when inherent hazardous agents or the pathogens they cause low consequences when used maliciously. Most of biological agents included in these first two groups, one is non-pathogenic and 2nd low malicious use risk.

Examples:

Let's take an example of *Mycobacterium leprae* is an acid fast bacilli; they give strongly acid fast reactions when stained with carbolfuchsin. This *mycobacterium leprae* causes a disease known as leprosy, now here the organism grows very slowly, the generation time is about 30 days. This organism is not highly virulent. We treat the patient by giving multi drug treatment. So this microorganism is included in the Low Malicious use risk group.

3. Moderate malicious use risk (MMUR)

The 3rd one is Mild Malicious Use Risk group. These pathogens, microorganism or hazardous agents cannot be deployed as biological weapons. It causes low or moderate consequences when used maliciously and it causes low or moderate effects on the economy of any nation. Many current agents are included in MMUR.

Example:

The example is of *C. immitis*. It's a fungus; it causes disease which is commonly known as desert fever. Desert fever is apparently asymptomatic. And it cures without any treatment.

4. **High malicious use risk (HMUR)/extreme malicious use risk (EMUR)**

HMUR:

High Malicious Use Risk, it causes national and international consequences it means that these are the inherent hazardous agents that causes High consequences when used maliciously and they can be exploit as biological weapons. These leads to high casualties and high economic impact as well.

Example

Let's take an example of Bacillus anthracis that cause disease known as anthrax this bacteria is a gram positive, rod shaped, spore former..

EMUR-HMUR:

Now the last one is Extreme Malicious Use Risk, sometime EMUR is considered as High Malicious Use Risk. Because it also causes high risk when used maliciously. But they are not found in the nature. So here high security measures are required.

Example:

Let's take an example of small pox caused by a virus known as Veriola Major Virus. Soviet Union in the past they have used small pox as bioweapons. After that worldwide it is decided that small pox should be eradicated from world by vaccination. Because this Veriola virus is highly virulent. It is contagious means it can be transmitted. So they are totally eradicated from the world by simple use of vaccines.

Question:- Canada biosecurity legislation explain in detail 10

Answer:-

Canada biosecurity legislation

Development:

- public health agency of Canada
- Canadian food inspection agency
- guidelines for human and animal pathogens and toxins

Canada Biosecurity also deals with the health of humans, animals and plants and it also deals with the environmental safety issue. So for that two agencies are involves first one is the Public Health Agency of Canada: It deals with the health of human being that no pathogens should enter into the Canadian environment i.e. lethal for humans being. Second one is the Canadian Food Inspection Agency: It gives approval of genetically modified crops before entering into market. All these agencies giving guidelines for humans and pathogens and toxins.

Guidelines:

- used by researchers and lab workers
- facilities possessing, handling, storing or using such pathogens
- update risk-evidence -performance based approach

If we are talking about Canadian Biosecurity legislation that deal with laboratories work then there are some guidelines that must be adopted by the researchers and lab workers now the building must be fully equipped and it should be designed in such a way that pathogens should not be released from the lab. So All facilities equipments should be there for processing of pathogens and toxins. If we are dealing Biorisk 3 or 4 groups then pathogens must be handled in BSC. Must be proper storage procedures of these pathogens.

Update risk: As the risk increasing we have to update the Biosecurity legislations as well, so that we can make our performance better. And this is the basic objective of the Canadian Biosecurity legislation.

Question: How risk can be reduced through risk management? 5

Strategies:

To reduce risk strategies should be adopted .Established large policies and implement them. Design programmes and small projects is dealing with risky agents to transparent with sponsors. Small policies have been design for us subpopulation management principle applied to small policies.

- should be adopted
- implementation of large policies
- programs/projects-to be transparent with sponsors
- management principles applied to small policies

Risk Mangment:

This is structured approach that occurs in steps to identify access and control the risk, After identification we have to process in place minimize and control. We have to control adverse risk consequences materialize.

- structured approach
- occur in steps
- identify, assess and control risk
- processes in place to minimize and monitor risk
- control adverse risk consequences-materialize

Steps:

early consultation-identify needs and cost, deferring irreversible decisions-need more time to achieve objectives, pilot studies - more information about risk, design flexibility-modify-future needs, precautionary actions, transferring risk to private sector -----insurance, less use of leading edge technology,

undertaking site investigation, staging the project-review at different steps, abandoning the project - too risky.

Question: Japan legislation

- Japan ministry of health, labor and welfare
- two pillars of biosecurity
- surveillance of infection and infectious agents
- regulations of pathogen handling

Japan ministry of Health, Labor and Welfare is involved to prepare these legislations. Actually this ministry deals with the two pillars of Biosecurity one is Surveillance of Infection and Infectious Diseases, that how infectious agent should be processed, handled and stored. Now the 2nd pillar of Biosecurity which is designed by the Japan ministry of health labor and welfare one is the Regulation of Pathogens Handling. That there must be special establishment if the procedures and there must be proper regulations and implementation of those procedures as well.

Duties:

- screening of foods, human, vectors at the point of entry
- Japan ministry of agriculture, forest and fisheries
- health issues-animals and plants
- bioweapon-prohibition laws

Japan has created different ways they screen food, even human beings, vectors at point of entry into the Japan. So food vector or humans being screened whenever they enter to the Japan. The Japan ministry of agriculture forest and fisheries deals with health issues of animals and plants as well. Biosecurity legislation without proper control of bioweapons is incomplete. So Japan Biosecurity legislations, like the legislation of US and Canada, it also deals with bioweapons prohibition laws. Due to these laws bringing bioweapons in the state of Japan is strictly prohibited. So if we want to deal with the biopathogens that are lethal for the human being for research purpose then we have to take the special permission from the Japan ministry so that they can enter into the Japan state. Otherwise bioweapons which are used in the terrorism they are totally prohibited for entry into the Japan.

☺ Best Of Luck ☺

A 70 year meat eating person eats approx 7 cows and goats and chicken
Which is not fire option thy oil fuel oxygen

Which is harmful for human lead, mercury, gold and arsenic

Which not cause exposure to chemical inhalation ingestion needle stick
and lab coat contamination

Which is used as bioweapon mny anthrax lgaya hy option ni yad

Emergency plan is done by?

People access to information environment justice and all

Cybernetic means

To hack attacker are viruses worms and Torin horses and all

Folic acid with wheat this transgenic plant is used to cure?

Gmo in india are?

AFIA abbreviation?

1. AFIA..... American Feed Industry Association

2. Which is not an example of obstacle.... option a

a.lighting b. Multiple locks c. Fireproof safes d. Water sprinkle

3. All are example of contamination zone except.....option a

a. b. Sealed rooms c. Sealed containers d. Personal containers.

Forget the option but missing one was.

4. Who proves Aristotle wrong.... Thomos

5. Which is not fire.....Co2

6. Which is not the duty of biosaftey officer

.....

7. Cybernetics means.....governance

8. Ingestion of hazardous material except...

(Check option in topic 93)

9. GM crops increase food allergy by50%

10. Microbes inlikely to cause disease.... WHO risk group I

11. See Permission for food stuff topic 155 .

12. Folic acid with wheat to cure.....Anemia

Paper was repeated some mcqs are as under

1. food allergy increases 50%

2. AFIA abbreviation? American feed industry association

3. Biological spill kit:

- household bleach

- 70% alcohol

- spray bottle

Fire extinguisher

4 Which is not fire option oil fuel oxygen and heat

5 a suit laboratory personnel must wear positive pressure air protective suit bSI4

6 souls is “breathed in” to the human embryo on the 40 day after fertilization—Islam

7 Obstacles are except

1. multiple locks

2. fireproof safes

3. water sprinkles

4. Lighting

8 Internal transport is except

1. shipping

2. movement from / to restricted area within facility

3. involve commercial

4. carriers

9 diagnostic labs

BSI 3

10 treatment via gene therapy

1. Leukemia

2. Hemophilia

3. Down syndrome
4. cystic fibrosis
- 11 treatments via yoga
- Psychological disorder baki option bhol gy
- 12 bioweapon causes
- 13 license for export of GM food is provided by
- 14 Permission for food stuff is necessary for
 1. ingredients of food stuff
 2. additives
 3. processing aid
 4. Packaging material
- 15 Containment zone have all except
 1. Conference room
 2. sealed rooms,
 3. sealed containers
 4. personnel suits

mcqs

- 1.virtue of ethics is an idea of....pjilosphr
- 2.plagrism is.....
- 3.in high risk tb labs work with high conc. ofput works at high risk
- 4.national biosafety rules 2005 have constituted instituional biosaftey committe under rule....
- 5.in india GMO's are.....
- 6.AFIA stands for.....
- 7.emergency plans are designed by.....
- 8.risk framework assesment proped in
- 9.toll of allergies due to GM crop has increased up to.....
- 10.deferring irreversible decision result into.....
- 11.....is used as an obstacle
- 12.harmful effect of GMOs may damage.....

13. gene therapy is now acceptable.....
14. risk assessment scheme assesses potential hazards.....
15. component of internal transport.....
16. in xenotransplantation growth of.....
17. unrequired event which occurs suddenly.....

#mcqs:

1. ___ is required to import/export /sale/purchase genetically modified food
 - a. License from federal agency
 - b. License from institutional committee
 - c. License from PAEC
 - d. License from PCSIR
2. Gene therapy is not acceptable for
 - A. Down syndrome
 - B. Infertility
 - C. Cystic fibrosis
 - D. Leukemia
3. Risk assessment scheme assesses potential hazard
 - A. By comparing variables
 - B. Scoring system
 - C. Both a and b
 - D. None of these
4. which is not fire
 - A. Heat
 - B. Oxygen
 - C. Fuel
 - D. Oil
5. Indian GMO's are also called
Alien species
6. Function of biosafety collaborating center is

- A. Applied biosafety programs and things
 - B. Vaccine develop for AIDs
 - C. Risk function to decrease incidence of disease
 - D. Increase scientific output
7. Biosafety level 5 us for
- A. Organic biology
 - B. Synthetic biology
 - C. Inorganic biology
 - D. Iss may b inorganic likha tha
8. Favr savr tomato 1994 k tha ek yaad nahi
9. Animals used for workingband droughting are called
- A. Devils of droughts
 - B. Slave animals
 - C. Beastsbof burden
 - D. Beasts of importance
10. Which statement is correct for Irreversible deferring decisions
- Need more time to achieve objectives
- Best of luck and remember me in prayers plzz

VU ZOOLOGISTS

BT605 Final term Past Papers 2017 solved

1. What are the precautions made in BSL2 as compared to BSL1. 2

The major difference in BSL1 and BSL 2 workers which are working in BSL 2 in lab must have training for handling infectious material.

Special practices and equipment also required.

We restrict movement of people in the lab having BSL-2.

Then special bio containment required for this lab especially biosafety cabinet.

2. Physical Security elements? 3

Topic no 129 Physical security Elements

It not only deals with protection of people but Also deals with hardware network programmes as well as protection of data .Now physical elements are of **three types** these are **abstacals notification and recovery stage.Obstakles** placed in the way of attackers.**Survillence and notification systems.Methods to recover quikly from disasters.**

3.Equipment used in BSL3, write 5 points 5

Equipment:now here special equipments are required for example apart from personal equipment glass eye feel any shoe cover we also require **special vaccume line** msut be protected with HEPA filters.WE need **Ducted air ventilation syatem.In biosafty cabnet HEPA filters used same in ducted air ventilation system.**biosafty cabnent actually a chamber that has three filters 1 hepa filater intake air and 2 Hepa filter for exit air in order to dispersed away the contaminated air from occupied area.BSL 3 must be fully documented facility design that include decontamination of larger pieces of equipments any sort of operation any parameter regarding procedure must be documented prior to performing the procedure.

How risk can be reduced through risk management? 5

Write note on special practice in BSL2. 10

Discuss malicious risk groups and Its Types? 10

BT605

4,Biosecurity?3

US Legistlation:Biosecurity Bill 2014 and supported lagistlation were passed by government Now they become biosecurity act 2015.We apply this act in every field related to agriculture,Related to pharmaustical.Agriculture biosecurity department has made cooperation collaboration with 400 organisations.Biosecurity act of US consist of 630 pages.

Future aspect of this biosecurity act 2015:

Biosecurity act support biosecurity sytem in any age.Regardless of the advances in technology in scientific field like biotechnology. **Scientific advances and advises** help to make right decision.

Objectives:Was to create a modern and responsive legislation framework. Improves the underpinning process so create **the robust biosecurity system** that benefits everyone. Now this robust system prevent entry and establishment of invasive and exotic pests which are lethal

because these species not only harmful for the environment but also related to the safety of humans also depends upon economy.

5.BSL3?

Topic 85

BSL-3:It is applicable to diagnostic lab clinical research areas even production and teaching facility's, BSL3 include those agents that cause fetal disease and potentially hazardous disease in the human being because they appear in the environment in the form of aerosols and human being can inhale them as result enter in to the body through inhalation rout. microbiological practices should be performed in the biosafety cabnint .

Equipment:now here special equipment's are required for example apart from personal equipment galss eye shield and shoe cover we also require **special vacuum line** must be protected with HEPA filters.WE need **Ducted air ventilation syatem.In biosafty cabnet HEPA filters used same in ducted air ventilation system.**biosafety cabnent actually a chamber that has three filters 1 hepa filater intake air and 2 Hepa filter for exit air in order to dispersed away the contaminated air from occupied area.BSL 3 must be fully documented facility design that include decontamination of larger pieces of equipments any sort of operation any parameter regarding procedure must be documented prior to performing the procedure.

Topic no 86

6.BSL-4:

This is the highest containment level it include those agents that are dangerous exotic because they have ability to appear in the environment or contaminate environment in the form of aerosol .Aerosols are easily inhale by the workers who are working in the lab and can be easily infected by the infectious agents.These are dangerous because cause lethal disease inhumans no special vaccines are available no special treatment no specil precautionary measures can be taken.Here we have to take special precautionary measures for workers who are working in lab in order to prevent the formation o aerosol as much as possible now there is unknown risk for transmission.We **divided BSL-4 in two types**

1) **A cabinet laboratory: Manipulation of agent in BSC.** We can manipulate culture infectious material in a biosafety cabinet. Now biosafety cabinet that is satisfied and used according to the manufacturer detail they have double doors material that contain infectious agent first passes through autoclave for sterilization then disposed of biosafety cabinet. Now in autoclave sterilization occurs 21 degree centigrade for 15 to 20 minutes pressure 15 pounds. Biosafety cabinet must be designed if infectious material not sterilized in autoclave it must be removed carefully from biosafety cabinet. Never put sharp things in BSC. It is better to avoid such material. Now apart from HEPA filters which are used in filtration that intake air and excrete air gas and vapor decontamination unit in BSC. Now the worker who are using biosafety cabinet must wear glass eye shields then shoes cover once they enter into the lab they have followed all those personal equipments which are required for the lab BSL4. It is responsibility of lab manager to look at repair and maintenance of

BSC. 2ND TYPE: A suit laboratory: Where personnel must wear positive pressure air protective suit. BSC used for all microbiological procedures and decontamination of gloves and apparatuses during lab operations.

7. Risk management?

Topic No 151 Risk assessment or Risk management

Article 15 Annex 3 of Cartagena protocol set forth in national biosafety guidelines. National Biosafety committee will monitor all activities that are performed by researchers whether they are according to biosafety guidelines or not. Also check the license. **Auditing of risk assessment** in Pakistan we followed the risk assessment scheme in which we compare the variables second method is scoring system in which we check out that whether activities are at which risk level according to data we assess the risk but management is very important performs by three ways the first one is **auditing second one evaluation of risk management** and third one is **field trials** which is carried out case by case studies in a scientifically sound manner.

8. Measures required for TB lab?

Biosafety measures for TB lab: Microbacterium Tuberculosis is caused by Microbacterium Tuberculosis and is contagious means it can be transmitted from one individual to another simply by coughing in action.

In lab we have to follow special biosafety measures for handling specimen like sputum blood or cultures.

- Can affect the code of practice,
- equipment lab design
- health facilities,
- health surveillance training
- waste handling.

9. Biosecurity plan 3

Design Biosecurity plan

Biosecurity plan:

Written plan prevents introduction and spread of disease. It is written documentation or written plan that prevents introduction of pathogens into nation that affect any population or sub population even it can affect organizations. It not only deal production of pathogen in to the environment or organizations also prevents spread of diseases. We have to look daily operations procedures also deals with disinfection method important part of biosecurity plan. Biosecurity plan start with disinfection.

Responsibilities: Principle investigator of laboratory they're involved in making biosecurity plan in the proper implementation of that plan. Investigator also involved in the trainings of workers according to methodology biosafety or biosecurity. Then lab workers also responsible for designing biosecurity plan. They have to get special training techniques then they apply standard procedures in the laboratory and implement them routinely in case of any spillage disinfectant should be done properly and principle investigator should involved. Principle investigator cannot work alone he or work in collaboration with

responsible officials. Campus security staff for 24 hours related to the biosecurity of lab organization or workers. If we talk about management services electricity gas water services. Such people must know the nature of lab work as well.

Responsibilities of responsible officer Ro: responsible officer work with in collaboration with principle investigator for designing biosecurity. He must be contacted if biological agents lost or theft in laboratory because this is very serious issue pathogens used for terrorism. Responsible officer must contact with agencies if threat. training also compulsory for Responsible officer. principle investigator and RO conduct workshop and trainings.

Other aspects: Risk assessment not only confined in a laboratory also applicable field of agriculture when handling animals or food industry this is how we identify lethal pathogens. Different methodologies depends upon type of work which we are performing. Personal protection start from building structures facilities policies equipments and other things like sanitation part of physical protection. In personal protection we have to wear gloves face mask face shield even respirators if we are dealing the work group. **Pathogen accountability** : If a disease can spread epidemic or pandemic it is medical need to design vaccine we have to do basic research on vaccine gose toward manufacturing stage there is risk management assessment. **Emergency response**: in case of any spillage we must contact with RO.

10. Biohazard is classified into four elements name them write their categories, name their substance

5

Classification:

- Category A, UN 2814-infectious substances affecting humans
- Category A, UN 2900-infectious substances affecting animals
- Category B, UN 3373- biological substances transported.
- regulated medical waste, UN 3291- waste or reusable material

- derived from medical treatment/research

11. Bioweapons? And their threats and how they migrate 10

Topic NO 160 Effort To Mitigate Biological Threat

Whenever we use word biological threat term comes in our mind bioweapon or bioterrorism

Bio mean life and weapon means infectious agents that are used weapon against human being known as bioweapons.

Bioweapon: Any infectious agent that bacteria viruses used intentionally to cause harm to human being known as bioweapon other term that used for bioweapons is bioterrorism these both are synonyms. Now at national and international plane or programmes developed against the biowar defence planning of effective biowar defence very difficult task whether national or international level. It affects the nations and brings fear among public also affects the scientific community.

Defence Against Bioweapons: **International cooperation** we have to cooperate with international agencies to control bioweapons. Also **transfer of technology** that help to create bioweapons, we have to **Support national actions**. These three things are important that help to create defence against bioweapon.

BTWC: Now for bioterrorism and bioweapons not a new idea. The UN also used bioweapons against their enemies. Biological toxin weapon convention was developed in 1972 when there is spread of bioweapon. Now BTWC is worldwide accepted which was established in 1972. It is institutionally weak and implementation is ineffective without investigation. BTWC is institutionally weak.

Biological threats: Terrorist non state actors, misuse of technology. Bacteria virus can be taken from lab. National and international extremists act as biological threat. Sometimes international agencies are involved because of unemployment poverty they locally hire agents so frustration among people increases due to poverty and unemployment in developing countries.

Efforts to mitigate threats: ACCORDING TO BTWC ACT must be designation of focal point must be central implementation authority dealing with biosafety guidelines. Code of conduct for biologists researchers awareness regarding biosafety biosecurity bioweapons among public.

Pakistan efforts to mitigate biological threats Pakistan approach: From line of war on terror They not suffer financial loss also suffer from human loss every year. Bioweapons are not the part of security matrix. In dealing with bioweapons Pakistan has made tremendous progress

In 2005 new legislations have been introduced in Pakistan regarding biosecurity. Also improve their administrative measures risk assessment risk management NCGLs. National core group of life sciences was established in 2007. Biosecurity biosecurity must present in syllabus. Syllabus once design by committee finally review by HEC.

BPSA 2008: Pakistan biological safety association. Their purpose is to train the trainer who are seeking for biological certification they cover the module which are risk assessment management and also designing lab.

Efforts: National institute of health-WHO. they work with collaboration world health organization to control the diseases. Export control act 2004 also functional in Pakistan this is for reexport overexport and transshipment of goods technologies and equipments. Special actions taken against them if we not follow rules 14 years imprisonment or 5 million rs fine.

12. How immunization can be provided to fight against pathogens. 5

Topic no 107 Immunization of staff

This is for personal safety and safety of worker. Immunization is an example of active immunity which is very necessary if you want to do work in the laboratories. If you appoint any person in lab then you have to check the vaccination card whether they fulfill all the vaccinations included EPI programme or not also **discuss with workers. Vaccines are** available for hepatitis B that should given to the workers very important virus in Pakistan Mutation not cause by immunological genetic factor but also cause by environmental factors microorganism is important part of environmental factors they bring about the mutation and cervical cancer is one of major cancer 80% female suffer from cervical cancer for that HPV vaccination available vaccination available for human papilloma virus it can cause cervical cancer in female we can get vaccination from age of 19 year and onwards .If exposure in lab then **therapeutic drugs** should be available in lab in order to treat workers.

13.Name the things for protection in first aid kit

Contains??:

- first aid box
- protective clothing
- safety equipment
- eye irrigation

First aid box:

- made up of material keep
the content dust
and damp free
- white cross with
green box
- prominent position
- easily recognizable

Box contains??:

- first aid manual
- bandages
- sterile dressings
- safety pins

14.Name lab safety things 2

Lab safety

- safety glasses,
- closed-toed shoes
- no food/drink
- long hair must be tied
- lab coat
- open flames-unattended

- flammable liquids
- skin contact-rinse off
- proper waste disposal
- ,liquid waste-labeled containers
- equipment must be cleaned and placed back,
- inform-chemical spill/thermometer breakage
- be careful-handling hot glassware and apparatus
- avoid taking excess amount of chemicals
- cell phones/ head phones are not allowed
- lab door must be kept close
- chairs not permitted -lab work is in session
- lab must be fully equipped with fire extinguisher
- fire blanket/safety shower
- eye wash/ first aid kit
- fume hoods/ sodium hydrogen carbonate

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Bt605

15.Cybernetics (2)

16.Elements of act?

17.Functions of technical adversary committee? 5

Function of technical Advisory Committee:1)examine applications researchrs about the project and recommend it to national biosafety committee for their approval they review control safety measures as well.Review research methodology as well.They also monitor commercial release of GM o and products into the environment.Look at safety regarding humans and environment as well.Provide information regarding national biosafety committee about approved projects upto which level project have been completed funds utilized .Also supervise implementations of erms and conditions

18.First aid equipment's? 5

Protective
equipment:

- mouth piece for
mouth-to-mouth
resuscitation
- gloves-protections
- clean up kit for blood
spills

19.Discus transport security element.

Topic No 135 Transport security Elements

It consist of two elements one is **internal transport** other is **external transport**.

Internal transport: It is transport of movement of biological materials with in the country.Movement of biological material from to restricted area but remain with in the facility.If we are moving biological material with in the institution we will refer reliable lab works but if we moving biological material from one city to another we will divide the area into three **groups one shipping area:** It is the area where disposing biological materials for transport here use your id card special documents regarding composition of **pathogens** 2nd **receiving area:**Here the receiving name should be mentioned he or she will bring id card and 3rd one is **disposal area:**That is incenrator used for pathogens and toxins.

External transport:

It is the movement of biological material to another country these are two elements of transport biosecurity. Movement of biological material from one facility to another. So this involves commercial carriers. More than 30 commercial carriers available most commonly used is Airta and WHO here able to move frozen samples material. External transport is expensive as compared to internal. **In external transport we divide material into two groups first group is category A second one is category B infectious material** or pathogens in the form of culture isolated DNA isolate RNA it is recruited in category B such material **required triple packaging system triple** packaging names indicates primary secondary or tertiary. **Primary layer** consist of specimen that protects the specimen any sort of leakage **second layer** protect the primary layer and water proof and **third layer** protect the secondary layer.

20. Write some activities of bio safety officer.

Activities:

- **biosafety consultation**
- **biosecurity consultation**
- **technical compliance consultation**

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Eutrophication

BT605-Biosafety & Bioethics (15-08-2017)

Total Q 45 36 Mcqs 5 short Q (2 and 3 marks)

21. Functions of physical security?

First component of biosecurity it gives not only protection to people working in the lab .It also protect hardware or the programmes with which organization the lab work dealing with then data and network.

Physical events and physical circumstances that can occurs it can occurs in the form of terrorism then natural disasters in the form of fires or explosive chemical physical security not only deal with protection of people lab workers. In order to deal with the physical events that is of terrorism and disasters be used the component of biosecurity that is physical security.

22. What are surveillance elements?

Surveillance/notification system:

- heat detector
- smoke detector
- lighting
- alarms
- cameras

Surveillance and notification system:

- These are the **heat detectors** Heat detectors can give us alarms there is high temperatures for samples as well as equipment then
 - **Smoke detectors** detect the smoke it protect the building from fire.
 - **Light** is proper in lab especially bio work then emergency alarms should be there
 - **Cameras** to control the access of authorized people to the lab.
5. Duties of technical committee?
 6. **Committee** 3rd Technical biosafety committee not only deals with biosafety rules also deals with biosecurity rules depending upon field which Gmo is used.

23. What are malicious risk groups, explain it? (10)

Topic no 125 Characterize Risk

Here we characterize risk according to the **Malicious Risk Group** : Non pathogenic 2nd one is low malicious use risk which is commonly known as LUMER). 3rd one is moderate malicious use risk MMUR.

4th High malicious use risk HUMER or extreme malicious use risk EMUR. These are five malicious risk group. Noe discussed them one by one.

1) Non pathogenic: These are inherent hazardous agents that cause no or insignificant consequences. These are non pathogenic not harmful for human beings. **LMUR IS SO CALLED**

low when inherent inherent hazardous causes low consequences when use when use maliciously. Most of biological agents included in these first two one is non pathogenic and **2nd low malicious use risk.**

Examples: **Mycobacterium leporae** is acid fast bacilli they gives acid fast reactions when stained with carbol fuchsin size of this bacteria are parallel nad arrangements are usually parisac this mycobacterium tubercloses cause disease lapraseorganism grow very slowly about 30 days. This organism is not highly virulent. We trat ateint multi drug treatment.

MMUR: These pathogens microorganism and hazardous material can cannot deploy as biological weapons .Low moderate consequences and low and moderate economic impact any nation. Many current agent evaluated as MMUR.

Example: c immuitis Funjus. it cause disease desert fever. No symptoms appears. Cure withut treatmet. bosaftay level 3MMUR. **HMUR:** national and international consequences it means that these are the inherent hazardous agents. High economic impact. High casualties **EXA**mple bacillus anthracis that cause diseae known as anthrax this bacteria gram positive rod spore fromer. In 2004 united sate of America spores from one person to another in the from of mailed letter ,mailed letters contain spores of anthrax.

EMUR-HMUR: it also cause high risk mailiciousy not found in the nature high security measures required. It should radiacted from the nature. If you have prepared genetically engineerd agents. Example: small pox. caused by **viruse veriola** major viruse sowiat union in the past they have use **small pox as** bioweapons worldwide decided small pox eradicated from world by vaccine .It is contagious tranfered from one person to **another must be eradicated by vaccine** now it has been done

24. How risk can be reduced through risk management? 5

Topic 126 risk reduction

Strategies: to reduce risk strategies should be adopted .Established large policies and implement them.

Desgin progrmmaes and small projects is dealing with risky agents to transparent with sponsres. Small policies have been design for us subpopulation management principle applied to small policies.

Risk Management: This is a structured approach that occurs in **steps to identify, assess and control** the risk. **After identification we have to process, in place, minimize and control.** We have to **control adverse risk consequences** materialize.

Steps: We have to **identify our needs, our requirements also** we have to look at cost. Defer in an irreversible decision we need some more time develop alternative ways to achieve goals. Need **more time to achieve objectives.** **Pilot studies:** purpose of these small studies is to more and more information about risk. **We have to design such a strategies in risk assessment methodology** that are flexible. Design flexibility modify future needs. **Precautionary actions** for biosafety system. Transferring

Risk to the private sector in the form **of insurance we transfer risk to** the insurance company. **Less use of leading edge technology we have to look at** the site investigation. Staggering the **project review at different** steps. Then it is **better to stop the project when it is too risky for human being for animals, plants as well as environment.**

25. Genetic modification

29: Ethics and genetic modification

Genetic modification:

- any alteration of genetic material
- capable of producing new substances
- performing new functions

26. Test tube baby? (5)

BSL 1 Laboratory facilities: (2)

- ☐ ☐ Lab doors must be kept control and there must be proper doors for access control.
- ☐ ☐ There should be deep sink for hand washing.
- ☐ ☐ Bench top should be made with a material that must be resistant to chemicals to organic

solvents and inorganic solvent.

□□ Similarly chairs are made up of such a material that we can wash them easily even disinfect them.

□□ Lab windows that open external air must be fitted with screens

2) Surveillance: (2)

These are the **heat detectors** heat detectors can give us alarms there is high temperatures for samples as well as equipments. Then **smoke detectors** detect the smoke it protect the building from fire. **Light** is proper in lab especially in the bench work. Then **emergency alarms** should be there, there should be **cameras** to control the access of unauthorized people to the lab.

Physical security:

Physical security that not only deals with protection of people but it also deals with hardware network programs as well as protection of data.

Now physical elements are of **three types** these are **Obstacles, Notification System and Recovery Stage.**

3) Biosecurity (3)

Biosecurity policies have to be established and developed. All the conflicts **between biosafety and biosafety** have to be resolved so the **good lab biosecurity system can be enforced and leads to the strengthen of biosafety system.** **Security and precautionary measures should be routine part of lab work**

4) Activities of human change fresh water: (2)

Human activities change clean water into waste water. It can be domestic activity; it can be Agricultural, commercial, industrial, storm water, or it can be due the runoff water

5) Impact of bioweapon:

6) How Radiation effect minimized

7) What are malicious risk group and what are three abbreviation: (10)

Here we characterize risk according to the **Malicious Risk Group:**

The 1st one is the Non pathogenic,

The 2nd one is Low Malicious Use Risk group which is known as LUMER

The 3rd one is Moderate Malicious Use Risk group known as MMUR.

The 4th one is High Malicious Use Risk group known as HUMER

The 5th one is Extreme Malicious Use Risk group known as EMUR.

These are five malicious risk groups. Now discussed them one by one.

1) Non pathogenic:

These are inherent hazardous agents or pathogens that cause no or insignificance consequences. These are non pathogenic organisms that are not harmful for human beings.

2) Then the 2nd includes Low Malicious Use Risk **LMUR IS SO CALLED low** because when inherent hazardous agents or the pathogens they causes low consequences when use when use maliciously. Most of biological agents included in these first two groups, one is non pathogenic and **2nd low malicious use risk.**

Examples:

Let's take an example of Mycobacterium leprae is a acid fast bacilli they gives strongly acid fast reactions when stained with carbol fuchshion. The size of this bacteria are parallel and the ends are rounded the arrangements are usually parisac.

3) MMUR:

The 3rd one is Mild Malicious Use Risk group. These pathogens, microorganism or hazardous **agents cannot be deployed as biological weapons.** It causes low or moderate consequences when used maliciously and it causes low or moderate effects on the economy of any nation. Many current agent are included in MMUR.

Example:

The example is of C. immuitus. It's a fungus; it causes disease which is commonly known as desert fever. Desert fever apparently asymptomatic. And it cures without any treatment. But some life threatening cases have been reported, so it is included in the biosafety level 3 and C. immuitus is considered as an example of MMUR.

4) HMUR:

High Malicious Use Risk, it causes national and international consequences it means that these are the inherent hazardous agents that causes High consequences when used maliciously and they can be exploit as biological weapons. These leads to high casualties and high economic impact as well.

Example

Let's take an example of Bacillus anthracis that cause disease known as anthrax this bacteria is a gram positive, rod shaped, spore former. **In 2004 in Maryland USA** these spores are transfer from one person to another in the form of mailed letters, as the mailed letters contain the spores of Bacillus anthrax.

5) EMUR-HMUR:

Now the last one is Extreme Malicious Use Risk, sometime EMUR is considered as High Malicious Use Risk. Because it also causes **high risk when used maliciously. But they are not found in the nature.** So here high security measures are required. And it should be eradicated from the nature. E.g. if you have prepared genetically engineered agents that is considered in the risk group of EMUR. Then it should be eradicated from the lab as well as from the experimental procedures.

Example:

Let's take an example of small pox caused by a virus known as **Veriola Major Virus.** Soviet Union in the past they have used **small pox as bioweapons.**

8) Application of biosafety rule:

National biosafety rule 2005 are applicable to **manufacturing of genetically modified product** for example **national biosafety rule 2005 has divided into different committees; one is the institutional biosafety committee, 2nd one national biosafety committee, 3rd one is technical advisory committee**

The applications of national biosafety rules are divided among these committees they hold the funds they give projects to institution. And if they want to manufacture a product from the institution by using rDNA technology then here national biosafety rules are applicable

Storage of microorganism to avoid hazards of bioweapons national biosafety rule 2005 is applicable. If you are using a product of gene technology for research purposes or your project deals with genetically modified organisms such as GM plants, GM animals and you are performing their **field trial then national biosafety rule 2005 is applicable.**

If you prepared a product from rDNA and you want to **export or import it again you have to follow the rules.** Even for the sales and purchase of GMO we have to follow the national biosafety rule 2005

9) Japan biosecurity legislations:

Introduction:

- **Japan ministry of health, labor and welfare**
- **two pillars of biosecurity**
- **surveillance of infection and infectious agents**
- **regulations of pathogen handling**

Duties:

- **screening of foods, human, vectors at the point of entry**
- **Japan ministry of agriculture, forest and fisheries**
- **health issues-animals and plants**
- **bioweapon prohibition**

Laws

10) Objective of lab biosecurity:

Those laboratories which are dealing with the risk group 3 or 4 they mainly design Biosecurity objectives. The 1st lab Biosecurity objective is to design the safety agenda

This support biosafety agenda to prevent disease. It is not only to safe environment but also prevents the spread of pathogens to the humans being animals and plants that ensure containment of infectious material. For this reason responsibilities are increased on the lab investigators **Biosecurity Officer they must look at Biosecurity** system of lab on routine basis.

The 2nd main objective of the lab Biosecurity is to the **Maintain citizen confidence on bioscience research** community. Now you have designed the product in the lab that is very useful if we market it, so it should be your experiment your product composition should be transparent to the investors in the industry.

Then the last but very important objective of the lab Biosecurity is to protect the risk of crime any sort of crime and even of bioterrorism, there should no release of bioweapons in environment. For that there **must be proper transport guidelines available** by AITA and WHO.

11) Surveillance elements: (2)

Surveillance and notification system:

These are the **heat detectors** heat detectors can give us alarms there is high temperatures for samples as well as equipments. Then **smoke detectors** detect the smoke it protect the building from fire. **Light** is proper in lab especially in the bench work. Then **emergency alarms** should be there, there should be **cameras** to control the access of unauthorized people to the lab

12) Biosecurity plan: (3)

[Design biosecurity plan](#)

- **written plan-prevent the introduction and spread of disease to farm**

- **daily operation procedures**

- **disinfecting procedures-part of the plan**

[Responsibilities:](#)

- **principle investigator plan implementationworkers following the plan-training**

- lab workers
- responsible official
- campus security staff
- management services

RO:

- contacted if biological agent is theft or lost
- contacted agencies if there is threat or spill
- training

Other aspects:

- risk assessment
- physical protection
- personnel protection
- pathogen accountability
- emergency response

13) Threats of bioweapon: (3)

14) Equipment in 1st aid: (5)

First medical treatment at the time and place of accident known as First aid.

This is approved method internationally. It is used to treat a **casualty** in case of accident before we handing over casualty and patient to the doctor before doctor treatment

Contain:

First aid must contain aid box, protective clothing (contain gloves, eye shield, face mask) safety equipment and eye irrigation. Safety equipment can be in the form of rope for fire hazards, mouth to mouth pipe for resuscitation then eye irrigations system must be properly placed in the first aid box.

First aid Box:

It is a special box made up of such a material that keep the contents dust and damp free. It has many symbols in different countries **white cross with green box** in some countries balck and some countries red and white. So symbol is varying depending upon the country in which we are using the first aid box. If we are dealing with laboratories any sort of accident occurs then first aid box placed in **prominent position and easily recognizable because of their symbol. Must have first aid manuals. It should contain simple things like cotton, paper towel in order to clean spillage, the patient bandages of different sizes like triangular and Sterilize dressing and safety pins safety pins also in box for dressing.**

Protective equipments:

It includes Mouth piece for mouth to mouth resuscitation then gloves are Important in order to protect hands from carrying infectious material. Clean up kit for blood spills or

any container spillage, there must be a disinfectant.

15) Functions of technical committee: (5)

Function of Technical Advisory Committee:

Is to examine applications researchers about the project, proposals and recommend it to national biosafety committee for their approval.

They review the safety measures as well, and they also control the safety measure after the approval of the project.

They Review the research methodology as well.

They also monitor commercial release of GMO and products into the environment.

They look at safety regarding humans health and environment as well.

They provide information regarding the national biosafety committee about approved projects up

to which level project have been completed, the funds has been utilized, whether the article has published in international journal or not.

They also supervise the implementations of terms and conditions. Because whenever an application is approved or a proposal project is approved it is always following the terms and conditions.

16) Risk reduction by management: (5)

To **reduce risk strategies** should be adopted. For example for large populations we have to **established large policies** and implement them.

Risk Management:

Now for risk reduction the term risk management is used. This is structured approach that occurs in **steps to identify access and control** the risk. **After identification of the risk we have to process them in a place where we can minimize and control the risk. We have to control the after effects of the risk and they should** materialize

Steps:

So the risk management that occurs in steps; the 1st step is early concentration.

For example if we have designed a lab then we have to **identify our needs our requirements and also** we have to look at cost.

The 2nd step is the Deferring irreversible decision we need some more time develop alternative ways in order to achieve our goals. So this is again a better approach to take more **time to develop different techniques to achieve the objectives.**

The 3rd one is the **Pilot studies**: these are small studies; the purpose of these small studies is to get more and more information about risk.

Then the 4th one is that we **have to design such a strategies in risk assessment methodology** that are flexible. So if the method is Design in such a way that we can easily modify them according to the future challenges.

5th one is the **Precautionary actions** should be the part of the daily biosafety and biosecurity system. E.g. if a technician transferring samples from collection center to the lab, and blood spillage occur in the way, so to manage this risk we need to take precautionary measures that is

procedure if disinfection.

Then the next 6th step is **Transferring of Risk to the private sector**; in the form of **insurance we can transfer the risk to** the insurance companies. Now it is better to use simple technique instead of using complex techniques. So less **use of leading edge technology**.

Then in 7th step we have to look at the site investigation.

17)

Which additional facilities in BSL4 as compared to BSL3:

1) **A cabinet laboratory:**

Manipulation of agent in BSC. be can manipulate culture infectious material in a biosafety cabinet

BSC. 2ND TYPE: A suit laboratory:

Where personnel must wear positive pressure air protective suit. BSC used for all microbiological procedures and decontamination of gloves and apparatus during lab operations.

18)

What are bio weapons, biological threats, and explain the effort to mitigate them;

19)

BT605 Final Term 2018 MCQS and Papers

Which of the following is not an advantage of patenting the human gene?

- a. Patents can help reduce the price of resultant products/medicine
- b. Synthesis of generic drugs- AIDS
- c. Genetic based treatments
- d. Gene patenting can fund the developing drugs.

2) Which of the following cannot be predicted with the help of genetics?

- a. Predict what disorder a person is more likely to develop
- b. Predict response of drugs in particular individual
- c. Predict the psychology of a person
- d. Predict the metabolic rate of an individual

3) Which one of the following is true for the state of completion of human genome project?

- a. 99% euchromatic human genome sequenced

b. 99.99% euchromatic human genome sequenced

c. Euchromatic regions are not sequenced

d. 99% heterochromatic regions are sequenced

4) According to Islam soul is breathed into human embryo on the

a. 50 days after fertilization

b. 40 days after fertilization

c. 30 days after fertilization

d. Just after fertilization

5) Which of the following is an inadvertent and unwanted effect of GM crops?

a. More productive crops

b. More nutritious crops

c. Killing of the bees and butterflies

d. Pharmacrops

6) One of the following is not a socio-economic argument related to genetic technology?

a. Inequalities in ownership of the technology

b. Exploitation of the poor by rich

c. Increasing sexually transmitted diseases

d. Act as lightning conductor for attracting opposition

7) First animal to be cloned was

a. Sheep

b. Frog

c. Sea urchin

d. Cattle

8) Children who are genetically engineered in the womb to have desired qualities are

a. Choreographed babies

b. Designed babies

c. Designer babies

d. Desired babies

9) In one prenatal diagnosis, a pregnant mother has found out that her baby is having Down's syndrome. All of the following are coping mechanism you will follow while advising the expecting parents, EXCEPT one?

a. Focus on the child's overall wellbeing

b. Provide unrealistic expectations for the future and models for coping

c. Explain conditions in an understandable way

d. Coping with stress of caring

10) DNA has the parts except

a. De-oxyribose sugar

b. Phosphate group

c. Nitrogenous base

d. Ribose sugar

11) Tuberculosis is a dangerous disease and able to transmit from one person to another.

Such diseases are called

a. Contagion diseases

b. Contagious diseases

c. Transmissible diseases

d. Syndromes

12) Moderate hazards effective to personnel and environment are contained in

a. BSL1

b. BSL2

c. BSL3

d. BSL4

13) Dangerous exotic agents are contained in

a. BSL1

b. BSL2

c. BSL3

d. BSL4

14) On BSL1 level of safety while working/ handling equipments, gloves are

a. Not necessary

b. Still mandatory

c. Should be worn before entering the lab

d. Prohibited

15) Biosafety level at which potentially lethal disease causing microorganisms through

inhalation route are cultured and applicable to

diagnostics/clinical/research/production/teaching/teaching facilities.

a. BSL1

b. BSL2

c. BSL3

d. BSL4

16) According to moral theory solution of a problem is to be

a. Encouraging

b. Against

c. Convincing

d. Discouraging

17) In gene editing DNA is

a. Inserted

- b. Deleted
- c. Replaced
- d. All given

18) ----- % dietary DNA is degraded by enzymes.

- a. 96
- b. 98
- c. 76
- d. 85

19) ----- is a greenhouse gas.

- a. CO₂
- b. O₂
- c. I₂
- d. F₂

20) In slaughtering Muslims and ----- cut the neck without stunning.

- a. Americans
- b. Jewish
- c. British
- d. All above

21) Human genome project has ----- % accuracy.

- a. 96
- b. 99.99
- c. 100
- d. 88

22) In nanotechnology ----- can live on a chip device.

- a. Heart cells

- b. Neurons
- c. Stem cells
- d. Kidney cells

23) Permission before getting the healthcare is ----- right of patient.

- a. Social
- b. Ethical
- c. Necessary
- d. Moral

24) Spread of radioactive isotopes is the accidental activity that causes

- a. Ozone depletion
- b. Soil pollution
- c. Water pollution
- d. Industrial pollution

25) Applications of biotechnology are related to

- a. Health
- b. Medicine
- c. Food and agriculture
- d. All given

26) CO₂ traps ----- radiations from sun.

- a. Beta
- b. Gama
- c. Infrared
- d. Ultraviolet

27) Cross pollination contaminate ----- crops.

- a. Annual

- b. Regular
- c. Seasonal
- d. All given

28) Most mothers accept the risk to their ----- health.

- a. Own
- b. Fetus
- c. Husband
- d. All

29) Animals can be used in ----- and fashion.

- a. Sports
- b. Companionship
- c. Leisure
- d. All given

30) Use of ----- as vectors increases the risk of cancer induction.

- a. Fungi
- b. Bacteria
- c. Virus
- d. Plasmids

31) It is the duty of physician that he/she must give complete ----- to the patient.

- a. Cure
- b. Rest
- c. Information
- d. Money

32) In patient physician relationship ----- has right to make decisions.

- a. Doctor

- b. Family
- c. Organization
- d. Patient

33) Soil degradation takes place if there is less number of ----- on land.

- a. Animals
- b. Trees
- c. Plants
- d. Humans

34) Nano medicines have ----- side effects as compared to chemotherapy.

- a. Less
- b. More
- c. No
- d. All given

35) There is 99% homology between human genome and the ----- genome.

- a. Monkey
- b. Mouse
- c. Cat
- d. Fish

36) Latest biomedical research has created -----

- a. Super humans
- b. Super horses
- c. Super bacteria
- d. Super viruses

37) Study of interactions between man, machine and animals is

- a. Cybernetics

- b. Bioinformatics
- c. Computing biology
- d. All given

38) Genetic modification is

- a. Alteration of genetic material
- b. Capable of producing new substances
- c. Performing new functions
- d. All given

39) Viral vectors lack ----- genes.

- a. Replication
- b. Resistance
- c. Expression
- d. All given

40) Emergency services include -----

- a. Address/Phone numbers
- b. First aid
- c. Emergency equipment
- d. All given

41) Hazardous chemicals follow the ----- route of exposure.

- a. Ingestion
- b. Inhalation
- c. Needle sticks
- d. All given

42) Transport of infectious materials require -----

- a. International regulations

- b. Packaging
- c. IATA shipping
- d. All given

43) Work with high concentration of ----- is the risk factor in high risk TB labs.

- a. E.Coli
- b. Bacilli
- c. Cocci
- d. Spirilla

44) Bio safety cabinets are used not only to protect the worker but also

- a. Prevent the lab from overheating
- b. Protect the lab environment from microorganisms
- c. Absolve off the lab workers from wearing lab coats
- d. Enable eating and drinking in the lab

45) If equipment are unnecessarily switched on this will lead to -----

- a. Fire
- b. Electric current
- c. Heating
- d. All given

46) If a person ingested a hazardous material then ----- is first step should be taken in this emergency.

- a. Identification of ingested material
- b. Treatment
- c. Complete record
- d. Removal of ingested material

47) In moderate risk TB labs there is ----- concentration of infectious particles.

- a. Low
- b. High
- c. Moderate
- d. All given

48) Personal health information is disclosed to family when patient suffer

- a. Infectious diseases

49) Favr savr tomato (1994) genetically has

- a. Blocked ripening gene

50) Staphylococcus enterotoxin b is -----

- a. Toxin

51) Accident report should be prepared within ----- days

- a. 5

52) Following is not component of act in human trafficking?

- a. Abuse of power

53) Cybernetics is Greek word which means

- a. Governance

54) Meta ethics is -----

- a. Theoretical

55) Which forest is called climax forest?

- a. Tropical rain forest

56) Which of the following is not correct?

- a. Patient who is mentally unstable have no right

57) Shortly after spillage which is priority

- a. First aid to worker injury

58) Aristotle idea was rejected by

a. Thomas

59) Nurses profession consist of except

a. -----

60) In moderate risk TB labs there is ----- concentration of infectious particles.

a. Low

b. High

c. Moderate

d. All given

61) If the condition of patient is critical how much percentage of information can be shared with him/her?

a. 70-80%

62) If a mother refuses treatment what is response of physician?

a. Physician should do what is beneficial

63) The word ethics was first used by

a. Greece

64) Which metal is not poisonous?

a. Gold

65) We can take stem cells from embryo before

a. 5-7 Days

66) Science is based on

a. Observation and experiment

67) HJP was started in

a. 1990

68) Following are public point of views about research EXCEPT.

a. -----

69) Serious neuro-degenerative disease is caused at the age of

a. 40

70) Embryos don't have emotional, ----- or psychological properties to be treated as human.

a. Intellectual

71) Type of allergy due to fur of animals is hypersensitivity type

a. 1

72) Incidence of which disease can be reduced by adding folic acid to the wheat by transgenic techniques?

a. Night blindness

b. Thalassemia

c. Hodgkin Lymphoma

d. Anemia

73) Development of ethics occurred as

a. Discipline and institutionalization

74) Human embryonic stem cells are obtained from

a. Inner cell mass/Blastocyst

75) Which of the following is important in development of bioethics?

a. Bioethics as discipline and institutionalization

76) Earth was created how many billions years ago?

a. 4.5

77) Living will indicates

a. Treatment plan

78) Mouse genome was sequenced in

a. 2002

79) 21st century is called as

- a. Science
- b. Modern science
- c. Postmodernism
- d. All above

80) Ethics is under discussion about ----- years ago

- a. 3000

81) Frog was cloned in

- a. 1950

82) Acromegaly, cranial deformities and infertility can be treated by using ----- in nutrition.

- a. Zinc (Zn)

83) Father of environmental ethics

- a. Rolston

84) Meta ethics is not

- a. Observational
- b. Theoretical
- c. Experimental
- d. None of these given options

85) Virtue ethics is an idea of ----- philosophers.

- a. Latin
- b. Greek
- c. Pagan
- d. Arab

86) One of the following is not included natural world. Choose that option.

- a. Plants
- b. Invertebrates
- c. Air, soil, water
- d. Homo sapiens

87) One of the following is not a part of the ecosystem, choose that option.

- a. Water
- b. Air
- c. Soil
- d. Light

88) What is most important advice for the reviewer of a research grant proposal in the code of ethics?

- a. Best interpretations
- b. Summarize honestly
- c. Disclose financial resources
- d. Use manuscript in confidential way

89) Following are the rights of the patients, EXCEPT?

- a. Right of respect and privacy
- b. Right of information and communication
- c. Right of quality care
- d. Right to terminate his/her life

90) Surgical delivery is recommended in case of?

- a. Tuberculosis
- b. Alzheimer's disease
- c. Hepatitis
- d. Syphilis

91) Incidence of which disease can reduced by adding folic acid to the wheat by Transgenic techniques?

- a. Night blindness
- b. Thalassemia
- c. Hodgkin's lymphoma
- d. Anemia

92) What sort of fetal tissues is obtained while doing the prenatal genetic diagnosis?

- a. Amniotic fluid
- b. Allantoic fluid
- c. Fetal skin tissue
- d. Chorionic villus

93) Degranulation of ----- releases allergy causing chemicals called histamines.

- a. T cells
- b. B cells
- c. Monocytes
- d. Mast Cells

94) According to lab safety protocols, taking excess amount of chemicals in laboratory can result in to

- a. Loss to the lab
- b. Fire of the fellow colleagues
- c. Ill reputation of the scientists who is taking it
- d. Contamination of lab by fumes

95) To ensure safety measures in laboratory settings, Biosafety depends upon

- a. Lenient instructions to be observed
- b. Immense trust on the laboratory staff

c. Leisurely use of laboratory resources

d. Strict guidelines to follow

96) Human embryonic stem cells are used to

a. Treat cancer

b. Make vaccines

c. Make antibiotics

d. Generate organs

97) Dolly the sheep suffered from which of the following ailment?

a. Problem in urination

b. Inadequate sight

c. Problem in walking

d. Respiratory distress

98) Religious thinkers believe that genetics screening is sort of

a. Playing DNA

b. Playing humans

c. Playing religion

d. Playing God

99) To be the subject of a patent, an object must be a/an

a. Discovery

b. Either invention or discovery

c. Neither invention nor discovery

d. Invention

100) Harvesting of human embryo violates the

a. Duty to prevent suffering

b. Both duties

c. Human embryo is not viable so it is not treated as human

d. Duty to respect the value of human life

101) In which of the following cases, would you grant permission based on ethical grounds?

a. The older sibling suffers from genetic disorder and the embryo created in vitro would be tested for the absence of mutation and is the positive tissue match to the older sibling

b. Normal fertile couple undergo in vitro fertilization in order to produce a baby that can be a stem cell donor for an older sibling

c. No permission to any one of these cases

d. The condition suffered by the older sibling is not genetic but the child still needs donated stem cells.

102) What is a carrier of a genetic disease?

a. The one who has the genetic disease and carry the disease to the next generation

b. The one who recovered from a genetic disease and carry the disease to the next generation

c. The one with symptoms and does not carry the disease to the next generation

d. The one not expressing the disease and without symptoms and carry the disease to the next generation

103) The process of designer babies can enhance the human life up to ----- years.

a. 40

b. 30

c. 20

d. 10

104) Human body contains ----- cells.

- a. 100 trillion
- b. 100 million
- c. 250 trillion
- d. 250 million

105) Human DNA differs between individuals by

- a. 0.1%
- b. 0.15%
- c. 0.2%
- d. 0.25%

106) Which of the following statement is NOT included in positive eugenics?

- a. Taxation of the childless
- b. Financial/selective incentives to have children
- c. Reduction of unplanned pregnancies
- d. Eugenic immigration

107) Biosafety levels were designed in USA by

- a. CSC (Center for Disease Control)

108) Hubert designed Biosafety cabinet in

- a. 1943

109) Biosafety levels are followed by European union but they are defined as

- a. Directives

110) Biological warfare labs were established in USA in (14 dedicates)

- a. 1955

111) American biological safety association (ABSA) was developed in

- a. 1984

112) Chemical fume hood is required for microorganisms.

- a. Identification
- b. Isolation
- c. Characterization
- d. All above

113) Only authorized persons can enter in the lab. Such labs are called

- a. Hot zones

114) According to WHO

- a. Mouth pipetting is prohibited

115) Lab windows that open externally must be fitted with

- a. Screens

116) To maintain the chemical composition of samples while transporting which is used

- a. Preservatives

117) Children and old people has immunity

- a. Weak

118) If there is chance of infectious aerosols, then it is recommended to use

- a. Biosafety cabinet

119) Biosafety cabinets has HEPA filters

- a. 3 filters (One for intake and two for exhaust)

120) Biosafety level 3 must be fully

- a. Documented

121) In autoclave sterilization occurs at

- a. 121C 15psi and 15-20mnts

122) In Biosafety measures for TB lab doors must have

a. Signs and symbols of biohazards

123) Mycobacterium can appear in the environment while centrifugation in the form of

a. Aerosol

124) If the materials are not sterilized by autoclave we use

a. Incinerators

125) The temperature of incinerator chambers are

a. 800 and 1000C

126) We can disinfect water using ----- g of chlorine per liter.

a. 50g

127) The first step for identification of mycobacterium is

a. Microscopy

128) The stain used for mycobacterium is

a. Niehl Zeelson

129) 2

nd step is fixation in staining. It maintains morphology but kills bacteria. There is chance of

a. Aerosol formation

130) Specimen-automated nucleic acid purification assay is used to ---- DNA

a. Purify

131) Vortex and shaking system are not allowed in

a. TB labs

132) Directional flow of air is required in

a. Low risk TB labs

133) In moderate risk TB labs there is concentration of bacilli

a. Low

134) The medium used to grow mycobacterium is

a. LB Agar

135) Inoculation and drug susceptibility testing has

a. Moderate risk of aerosol formation

136) If the patient is suffering from high level of TB then the sample should be marked

with ink

a. Red

137) Laboratory room is decontaminated with the help of

a. Defumigation

138) Biosafety cabinets protect

a. Workers and environment

139) The size of aerosol is ----- which can't be seen with naked eye.

a. 0.5um

140) Negative pressure flexible film isolators has inside pressure

a. Low

141) HEPA filter can trap microorganism having size minimum of

a. 0.3um

142) Open flame in Biosafety cabinet can interfere with

a. Ventilation system

143) Micro incinerators are made up of glass and ceramic and they are

a. Disposable

144) Vacuum line system is consisted of flask and rubber bulb. Most important is that

it is

a. Auto-claveable

145) Latex gloves can cause

- a. Type 1 hypersensitivity

146) Which protection is required where we are dealing with animals

- a. Eye and face

147) Respirators are needed when we are dealing with risk group

- a. 3 or 4

148) When worker has ingested the infected material then the first step is to

- a. Identify the material

149) In case of natural disasters we must have

- a. Phone numbers

150) Transgenic organisms never existed in

- a. Nature

151) First DNA technology was limited to

- a. Cloning DNA

152) Most common vector is

- a. pBR322

153) pUC18 is routinely used in labs because it is

- a. completely sequenced

154) EcoliK12 is non-pathogenic and occurs in

- a. Gut

155) In recombinant DNA technology we commonly use vector

- a. Viral vectors like adenovirus

156) Adenovirus vectors lack

- a. Replication genes

157) Which is routinely used in antibacterial solutions?

a. Azide

158) Picric acid or picrate can explode by

a. HEAT

159) Compressed gas must not be

a. Incinerated

160) Alpha rays, beta rays and x-rays are

a. Ionizing radiations

161) We use ionizing radiations to measure

a. Hormones (I-125)

162) Biosafety officer not only deals with laboratory but he/she must have knowledge of

a. Clinical practices

163) Biosafety training should be done every

a. 3-6 month

164) Oven temperature must be restricted to

a. 80C

165) Emergency plans are designed by

a. Biosafety officer

166) If chemicals are in the form of crystals then they must be stored at

a. 4C

167) If a worker has suffered from chicken pox then it may happen

a. Never in life

168) 80% of females are suffered from

a. Cervical cancer

169) HBV vaccines can be given to age

a. Any age

170) Vaccination for human papillomavirus can be given after age

a. 9 years

171) Biosecurity act 2015 of agriculture department USA is consisted of

a. 630 pages

172) AFIA provides guidelines for

a. Bioweapons

173) Whenever we want to develop a biosecurity plan we look at

a. Location

b. Business

c. Facility

d. All above

174) Bio-risk management is the responsibility of

a. Management director

175) Hazardous substance and new organism act was developed in new Zealand in

a. 1996

176) Biosecurity plan starts with

a. Disinfection

177) The number of international commercial carriers are

a. 30

178) For international transport, infectious materials are included in category

a. B

179) During past few decades 30 new pathogens have been discovered. Of the 30 pathogens 16 are

a. Viruses

180) Virology lab must be

- a. Three storey building
- b. End of corridor
- c. Isolated
- d. All above

181) Virology lab administration is consisted of

- a. Qualified virologist with 5 years experience
- b. Two junior microbiologists
- c. Two lab technologists
- d. All above

182) Micropipettes of all sizes are available EXCEPT

- a. 5
- b. 20
- c. 50
- d. 100

183) For agitation we use

- a. Shaker water bath

184) For every virology lab we follow guidelines for

- a. Biosafety level 3

185) In 1883 in England ----- children died of fire.

- a. 180

186) In 1911 in America ----- worker died out of fire.

- a. 146

187) Running green man as a sign of exit during fire was introduced by

- a. ISO 7010 in 2003

188) The assembly must be far from fire hazard

a. 50ft

189) Fire warden take updates from

a. Emergency crew

190) PARC stands for

a. Pakistan agricultural research council

191) Gastroenteritis is common in

a. Children

192) Chemicals in food like food color and meat can cause ----- and it came into knowledge in

a. Cancer , 1973

193) Stockholm convention is a framework risk assessment of

a. Chemicals

194) Risk assessment in sub-populations can be used when risk is

a. <1%

195) Toxins are ----- which revoke immune-----

a. Antigens , response

196) Toxinosis is pathogenecity caused by

a. Bacterial toxins

197) We can evaluate toxins and pathogens by

a. Serological methods

198) Virulence or virulent factor is assessed by

a. PCR

199) Receptor for BT toxin in humans is

a. Absent

200) Only transgenic fish that is acceptable commercially is

a. Zebra fish

201) Patients are unable to decide in disease like

a. Alzheimer

202) Worldwide patients of tuberculosis are 3.2 million, HIV Aids are 3.3 million while cancer patients are

a. 2.5 million

203) In lupus disease we can measure stress by

a. Anti-ribosomal antibodies

204) Legal status of fetus begins when it is

a. Viable

205) Hodgkin lymphoma can be treated by inserting genes in

a. Tobacco plant

206) In USA -----% food contain genetically modified ingredients.

a. 75%

207) Chronic myeloid leukemia is diagnosed by

a. Philadelphia chromosome

208) Silicon chips can function like

a. Nerves

209) Xenotransplants use organs of

a. Other species

210) Peter was a philosopher who gave the idea of

a. Animal rights

211) Dengue virus has serotypes

a. 4

212) Down syndrome is trisomy 21 while turner syndrome is trisomy

a. 18

213) Woferin is a

a. Blood thinner

214) Codeine is

a. Analgesic

215) More than 100 genes are present on chromosome

a. 6

216) Auto-immune diseases are genetic in nature and proteins responsible are present on

a. MHC (Major Histo Compatibility complex)

217) Phenyl ketone urea is due to changes in

a. Alanine

218) Vitamin A enhanced rice are called as

a. Golden rice

219) Average life span in Africa is less than

a. 40 years

220) Factor IX is a blood clot protein which is used to treat

a. Hemophilia

221) Third parent in designer babies is

a. Mitochondria donor

222) For surgical apparatus disinfectant used is

a. Glutaraldehyde

223) Formaldehyde is used as disinfectant for

a. Histopathological sections

224) Q fever is

a. Zoonotic disease

225) When Ige binds with mast cells ----- is released

a. Histamine

Solved Final papers 2018 of BT605 (Bioethics and Biosafety)

1) Explain science in 21st century.

In 21st century science is defined as "Investigation of universe by a set of methodologies". Taking observations, making hypothesis, testing hypothesis through experimentation and then concluding the hypothesis to increase observations is the theme or method of modern science. So there is a lot of space in this definition.

2) Name some issues which arise from or associated with scientific research.

- Animal concern
- Environmental concern
- Ethics related to Stem cells
- Ethics related to Cloning
- Participation of human subjects in research

3) Describe attitudes to science.

Does scientific research is valuable or not? This is the defining question for attitudes to science. Most people believe that scientific research is valuable and useful for us as well as for future generations. Few are against this belief. Let's take some examples

Neitzsche belief:

He said that Does God Dead? because people are deciding their own moral values.

Wittgenstein:

He was of the theory that scientific values must be interpreted in social context. His explanation seems more impressive and effective as the knowledge helps public to make their decisions proper.

Supporting attitudes:

Scientific research is helping in economic growth, public health, technological improvements, comfortable lives as well as quality of life.

Opposite attitudes:

Scientific results are not built experimentally but they are constructed socially. It's a great threat to scientific research.

4) What are principles of bioethics?

There are three principles of bioethics introduced by Toms and James.

1. Autonomy

- Autonomy is the right to decide about action by respondent in the form of informed consent. Physicians must deliver all pros and cons of the treatment given to the patients and must take them into confidence before start.

2. Beneficence

- It is the act of doing beneficial treatment only. The treatment must be beneficial to the patients.

3. Non-maleficence

- This principle is about doing no harm to the patients. It is difficult sometime to built a balance between beneficence and non-maleficence because of overlapping.

Example: Lupus diagnosis criteria must be followed to treat. International criteria available must be fulfilled before start.

5) Explain decision making in detail.

Ethics is about making decisions and selecting choices. Decision making in ethics has very long and complex history. Among Greeks it was believed that expressions are more effective than following rules. Making ethical choices requires strong ability to decide between competing options. There are 7 steps if followed can make efficient decisions.

1. Stop and think (Sana)

2. Clarify goals (Can)

3. Determine facts(Develop)

4. Develop options(Dominant)

5. Consider consequences (Carrier)

6. Choose (counseling)

7. Monitor and modify (Map)

6) Enlist four approaches which show the relationship of humans with nature.

1. Anthropocentrism (Humans are superior)

- This approach is based on human beings. Rest of the nature is for the good and benefit of humans. Humans can use anything they like or need without any regulation. This has already created too much destruction to ecosystem due to human greediness.

2. Ecocentrism (Ecosystem is superior like non-living)

- This approach is ecosystem centered. Ecosystem is consisted of non-living things like soil, water, forests and mountains. All living things depend on non-living things for their survival. Human misuse of ecosystem is in turn dangerous for humans. Humans are responsible for all biological life.

3. Biocentrism (Biosphere is central including humans)

- This approach focuses on biosphere. Biosphere is all living things on planet earth. Human is one of the living things. Humans can use resources freely until others are not harmed.

4. Theocentrism (God is creator)

- This approach is centered in the belief on God. Everything belongs to God. God is the creator and environment belongs to God.

7) Explain the difference between intrinsic and instrumental values. (5Marks)

Intrinsic Values Instrumental Values

1- Every living thing has its own rights or values plants, animals or natural resources.

1- Materialistic needs of humans are important which are fulfilled from natural resources.

2- It involves religion. 2-It does not consider religious teachings.

3- It emphasize on sustainable use of resources to maintain a balance.

4- This approach focuses on prohibition of use of living resources to give them respect and value.

5- The purpose of approach is to provide protection to the resources.

3-It focuses only on needs regardless of sustainability.

4-This approach ignores the feelings and pains of resources.

5-Purpose is the survival of humans at any cost.

8) What are themes in environmental ethics?

According to Rio-Declaration in 1992 there are two themes in environmental ethics

- Precautionary principle

- o It means we should think, plan and program before doing any action on environment. Its old concept of being careful to avoid unwanted losses.

- Sustainability

- o It is based on repeatedly using the resources not all at once to ensure sustainability and availability of resources for long time. Agriculture is the best example of sustainability

which is being repeated every year.

9) Explain effects of Global Warming.

- Climate zone is shifting
- Polar ice is melting
- Sea level is rising
- Metabolic rate of methane producing bacteria is increasing
- Species are becoming extinct

10) How human activities effect the biodiversity?

Humans are major threat to biodiversity. Pollution in the environment is the result of human activities either in the form of industries or deforestation. Biodiversity depends on ecosystem for survival. Any of the constituent of ecosystem like water, air, soil, climate and geography has direct effect on biodiversity. These factors are intensively disturbed by human activities which are leading to loss of biodiversity on planet.

11) What are some ethical issues of using animals in research?

- They have their own rights
- They bear pain and have feelings
- Killing results in disturbance of ecology
- Keeping animals in closed vicinities affect their behavior and make them depressed.
- They may transmit diseases to next generations which are being tested.

12) How animals are slaughtered in industries?

In industries the animals are pulled from legs and hanged in prongs of automatics conveyers. They are then passed through a container containing water in which only head is dipped in and current is passed. In this way they are stunned and then neck is cut to kill them.

13) Explain codes of ethics for biologists. (5marks)

1- Synopsis

- The summary of the work should be followed in spirit. What is written must be practiced and not anything undescribed.

2- Best interpretation of results

- After experimentations the results should be interpreted in honest and best way regardless of their accuracy. Similarly work of others must not be written with your own reference because it is plagiarism.

3- Summarize data honestly

- As biologists we should summarize data honestly.

4- Acknowledgement of contributors

- A biologist must acknowledge the participants or helping persons either biostatisticians or patients which played a role in the work.

5- It should be confidential

6- There should be no inaccurate or misleading data in the research thesis.

7- Financial supports resources must be acknowledged as well

8- He/she must provide help to colleagues or peers in their work.

14) What are codes of nurses?

Nurses have following four codes of ethics while performing duties.

1- Nurses should give proper care, response and respect to the patients in order to make the patients feel comfortable.

2- Nurses must have good practice and command on their practices like injections, looking after, punctuality and polite language.

3- Nurses should be professionally trained and equipped with required skills.

4- Nurses should cooperate with their co-workers in the hospital to ensure team work. They teach and train their coworkers as well respect them.

15) Why patients should cooperate with their caregivers?

It is the responsibility of the patients to cooperate with the caregiver to make sure the success of treatment plan. If they don't cooperate, then it would be very difficult for the physician to treat the patient in full spirits either to diagnose or to treat.

16) What is difference between objective bad news and subjective bad news?

Objective bad news Subjective bad news

Type of bad news which is related to objective data like disease information or treatment plan is called as objective bad news.

Type of bad news which is related to perception and response about objective bad news is called as subjective bad news.

It deals with information. It deals with emotions.

17) Explain types of patient's advance directives. (5marks)

Advance directives are needed when patients are unable to make decisions on their own. These are being practiced in various advanced countries but not in Pakistan.

There are two types of patient advance directives

1- Living will

- It is type of advance directive in which patients or guardians give their consent about future actions like if dialysis needed in future, should it be given or not. Similarly if patient need ventilator, then it must be given or not. Similarly any organ must be donated or not.

- Living will may be oral.

- Terminal illness is also a type of living will in which patient may die shortly and he/she can make immediate decisions about body parts or related.

2- Health care power of attorney

- The 2nd type of advance directives is power of attorney. In this type patients give full power to any agent or proxy to make decisions about him/her in future.
- It is also called as durable power of attorney.
- This directive is followed in case of complete unconsciousness of patients like comma or loss of nervous system functioning.

18) Why personal health information should be kept confidential?

Personal health information is secret or confidential information between patient and physician. It may have negative impacts on patients' life if information is leaked or shared. Although in some cases it is mandatory to share information for the sake of health of others but it may have following negative impacts on patients' life.

- It may lead people to hate the patient because of disease.
- It may affect his/her insurance policies.
- It may result in losing of job.
- It may direct patient to psychological distress among peers.
- Relatives may boycott the patient in fear of being affected.

19) What are aims of disaster management?

There are four aims of disaster management.

1. Assistance

a. Immediate or continuous assistance is required for people who have faced any disaster to maintain life.

2. Health improvement:

a. Disasters result in many injuries to the people, so aim is to help in treatment of injuries and diseases. For this medical camps are temporarily built to provide assistance.

3. Support the morale:

a. People who are victim of disasters are depressed and pessimist. Morale of the affectees should be raised with the help of media or personal efforts.

4. Limited aid:

a. To provide aid to the victims like transport, food, shelter for temporary bases and then on permanent bases is also the aim of disaster management.

20) What are patient's rights and responsibilities for refusal treatments?

Rights:

Patients have the right to decline the treatment. For example during chemotherapy a lot of pain is felt by patients. So he/she has the right to decline the treatment. Physician cannot force the patient to continue treatment plan.

Responsibility:

By declining treatment plan, patients should be informed completely about the consequences of decline. After knowing all the consequences patient has to take the responsibility on its own.

21) What are moral principles of dealing with human embryo?

Whenever we talk about human being research then two fundamental principles arises that are related to moral issues that whether it is good or bad.

1. Duty to prevent the suffering of individual:

2. Duty to respect the human life:

Both of these cannot be followed side by side or simultaneously. When we harvest human embryo we are violating second principle by not giving respect to the human life.

Now a day stem cell research is hot topic. Stem cells are very useful for therapeutic purposes. So if we use stem cells in research we are not respecting human life from which we have taken the cells.

Similarly if we wish to prevent the pain or suffering of individual animal, then 2nd duty of respecting life would be violated.

22) What is low risk in TB labs?

Low risk TB labs are so called because there is minimum requirement of Biosafety which are needed in

TB labs. The first step to identify the mycobacterium tuberculosis is simply microscopy. Ziehl-Neelsen staining is used in preparation of smear of M. tuberculosis.

Why low risk:

- Sputum is viscous and there are very less chances of aerosol formations.
- Simple staining is required.
- Microscopy is applied to identify the presence of Mycobacterium.

Precaution:

While air drying the sample, there may be chance of aerosol formation, so workers must be careful about.

23) What are ethical issues in using human embryonic stem cells for therapeutic purposes?

- Moral status of embryo----- destruction:
 - o For therapeutic cloning embryo has to be destroyed to obtain stem cells from blastocyst. This arises the issue of destruction of life of humans.
- Patient has the right to live:
 - o The patient suffering from any disorder which can only be treated using stem cells has the right to live also. So we have to decide about whether the patient is important or embryo.
- Potential donor exploitation:
 - o Females who are donating the embryo for therapeutic purpose suffer from ethical issue of being depressed or pessimist because she became pregnant and she has to give birth to his child.
- No access and benefit to poor communities:
 - o Therapeutic cloning is very costly. Poor cannot afford the expenses. So here ethical issue

of equality arises.

- It raises social and healthcare disparities.

24) Write three categories of biohazards related to animal use or handling.

While handling animals there are three categories of biohazards

1. Physical injuries

a. If the worker already has injuries then there is a chance of transmission of bacteria, fungi, parasites or viruses from animal to worker. Similarly if animals make scratches on skin or bite, then there is also chance of transmission of such pathogens which may prove lethal.

2. Zoonosis

a. The phenomena of transmission of diseases from animals to humans are termed as Zoonosis while the diseases are called as zoonotic diseases. For example Q-fever in cattle can be transmitted to humans because it is caused by bacteria. So care must be observed while handling animals.

3. Asthma and allergies

a. Animal fur, feces, urine and mucus are major causes of type 1 hypersensitivity also called as immediate hypersensitivity. Dust particles attached to animal fur may prove dangerous or infectious. Asthma is also included in the list.

25) What is sentience?

The capacity to experience pain and pleasure, experience satisfaction and dissatisfaction or capacity to experience comfort or discomfort is called as sentience.

26) Define biosecurity.

Preventive measure dealing with the life forms is biosecurity. More precisely it is a set of preventive measures used to reduce the risk of transmission of infectious agents from one organism to other or even from one state to another.

27) What are types of bio-weapons?

Bioweapons are

- Germ weapons
 - o Bacteria, fungi, recachesia, viruses
- Chemical weapons
- Radiological weapons
- Nuclear weapon

28) Explain exploitations in victim human trafficking.

Human trafficking is exploited or misused in following ways

- Prostitution
- Sexual exploitation
- Forced labor
- Slavery
- Organ removal

29) What are steps of gene patenting?

Genes only can be patented if they are invented. Discovered genes are not patented because they are natural or public property. The gene patenting has following steps

- Isolate gene from rest of DNA
- Make a copy of it from mRNA population
- Turn gene sequence into an invention
- Not a gene itself but a copy made in the test tube can be patented

30) What are cybernetic ethical issues? Enlist three

1. Machines are in charge of key human functions
2. Only wealthy ones can communicate through cybernetics

3. Senses and impulses may be transmitted in harmful way

31) Explain elements of human trafficking.

There are three elements of human trafficking.

1. Act

a. Recruitment

b. Transport

c. Transfer

d. Harboring

e. Receipt of person

2. Means

a. Threats

b. Use of force

c. Abduction

d. Miss use of power

e. Fraud

f. Payments or benefits

3. Purpose

a. Prostitution

b. Sexual exploitation

c. Labor

d. Slavery

e. Removal of organ

32) What is mass destruction weapon?

The weapon which can destroy everything at larger scale like life, buildings, dams or any other structure are called as mass destruction weapons. Germ weapons are mass destruction in the sense that they can only destroy living things like humans not buildings.

33) How genetic counselors design the session?

There are 5 steps which a genetic counselor follows to make a session.

1. Intake phase
2. Initial phase
3. Encounter phase
4. Summary phase
5. Follow up phase

34) Enlist and explain types of pathogens.

There are four types or classification of pathogens.

1. WHO risk group 1
 - a. Microbes unlikely to cause disease
2. WHO risk group 2
 - a. Microbes causing diseases but unlikely to be serious
3. WHO risk group 3
 - a. Pathogens causing serious diseases
4. WHO risk group 4
 - a. Pathogens causing serious diseases
 - b. Transmission
 - c. No treatment or prevention available

35) What are ethical issues in genetic screening?

Genetic screening has following major issues

- Religion
 - o Religious scholars say that it is playing with God. So it must not be done.
- Expensive
 - o Genetic screening test are costly and expensive. Poor cannot afford them.
- Common diseases__

o It is again debated that whether common diseases should be tested by genetic screening or simple biochemical tests.

· Medical record:

o It is also debated between different school of thoughts that whether genetic screening test results should be part of medical record or not.

· Social issues:

o Positive results of genetic screening can lead the patients to face social issues like hate, propulsion, being avoided by others, psychological issues, etc.

· Jobs or insurance:

o Genetic screening of a person may lead to losing of job or denial from insurance companies.

36) What are fields in which Biosafety is used?

Biosafety is used in following fields

· Ecology

· Agriculture

· Exobiology

· Synthetic biology

· Medicine

· Chemistry

37) What is lock box situation?

Lock box is a situation in which patient advice physicians not to disclose any information about their diseases, treatment plans or anything related even to their close circle relatives. But if the patient is suffering from infectious diseases then physician has to disclose information even in lock box to save others from being affected.

38) What are applications of cybernetics?

Applications of cybernetics are

- Limbs replacing wooden limbs
- Heart pacemakers
- Artificial retinas
- Silicon chips replacing lost neuronal functions
- University ID card Chips

39) What is advancement of health and medicine in biotechnology?

Applications of biotechnology in health and medicine are

- Vaccinology
- Diagnosis and gene therapy
- Xenotransplants
- Designer babies
- Genetically modified embryos

40) What are ethical issues in treatment of older siblings?

Older siblings suffering from life threatening diseases can be saved using stem cells from younger ones.

Ethical issues in this case study are whether it is good or bad to use younger child for the welfare of elder one or not. It is totally on behalf of the parents to decide. It is a very complicated situation. Only one sibling can be saved.

41) How human activity is responsible for loss of biodiversity. Enlist three factors.

1. Humans are cutting trees and clearing forests for land.
2. Pollution is resulting in loss of biodiversity.
3. Tropical rain forests are climax forests but they are being cleared at very rapid speed for various needs like wood, rubber,

42) Explain types of sterilization with examples.

There are two major types of sterilization

1. Physical sterilization

a. Heat, radiation, filtration

2. Chemical sterilization

a. Glutaraldehyde

b. Formaldehyde

c. Chlorine

43) Define cybernetics.

Study of interactions between man, machine and animals is called cybernetics. The word cybernetic is derived from Greek language which means governance. It explores regulatory systems with structures and functions.

44) Define Biosafety.

Prevention of large scale loss of biological integrity is called as Biosafety.

45) How biotechnology can be misused?

1. Reality of war

a. Biotechnological weapons can be used against enemies.

2. Inequality of resources:

a. Countries which cannot afford biotechnological achievements would be deprived of its fruits.

3. Continuous misuse:

a. If science is continuously misused it provides another angle of observations that our struggle or aspirations are not good all the time.

46) What precautions should be kept in mind while using centrifuge?

Centrifuges are present in every lab. So while using centrifuge we should keep in mind following precautions.

1. The centrifuge tubes must be used in pairs and equal volumes.
2. Equal number of centrifuge tubes should be placed in opposite to each other to create balance.
3. Make sure to seal the lid before starting the centrifuge.
4. In case of spillage don't be an engineer instead informs your lab supervisor or advisor about spillage.

47) What are two precautions to use thermometer?

1. Thermometers should not be used as stirring devices.
2. Bulb of thermometers should be kept away from flames because it can catch fire.

48) What are two types sterilization with respect to heat?

There are two types of sterilization with respect to heat.

1. Dry heat

a. Red heat, flaming, hot air oven

2. Moist heat

a. Boiling, steaming, autoclaving, pasteurization

49) What are properties of biohazard symbols?

1. It was developed in 1966 by Charles Badwin.
2. It is easy to sketch.
3. It is acceptable by all ethnic groups.
4. It can be recognized quickly.
5. Labeling of biologically hazardous materials become easy instead of writing a complete text of instructions just put symbol.

50) Explain classification of biohazards.

Biohazards are classified into four main categories.

1. Category A UN 2814

a. Infectious material which affect only humans

2. Category A UN2900

a. Infectious material which affect only animals

3. Category B UN3373

a. Biological material which is being transported like samples for diagnosis.

4. Medical waste UN3291

a. Waste or reusable material

51) Name at least 10 components of biological spill kit.

1. Household bleach

2. 70% alcohol

3. Sterilization bags

4. Yellow trash bags

5. Disposable gloves

6. Paper towels

7. Spray bottles

8. Surgical mask

9. Safety goggles, shoe covers, face masks

10. Lab coat

11. Metallic tongs

12. Procedure manuals for spills

52) What is informed consent? What are its elements?

Informed consent is permission before start of healthcare treatment or research. It is the legal and ethical right of the patient. It has following elements.

- Right to decision
- Information about risks and benefits
- Acceptance of interventions by patients
- Assessment of patient understanding

53) What are ethical issues of nanotechnology?

- Nanoparticles are highly toxic and reactive.
- They can penetrate in the environment polluting it.
- They can cross the membranes and can reside in the cells to harm.
- Nanotechnology may dislocate industries.
- Economically it can create imbalance.
- Privacy of patients is also disturbed.

54) Write names of any six agents of bioweapons.

1. Anthrax
2. Small pox
3. Enterotoxin B
4. Viral hemorrhage fever
5. Botulinum toxins
6. Brucellosis

55) What are elements of an accident report?

- What was nature of accident?
- Person involved in the accident
- Witness involved
- Causes of accident
- Signature of investigation officer

56) Explain BSL3 (Any 5 point).

- All procedures are performed in a biosafety cabinet 2.
- Three HEPA filters are present in cabinet, one for intake of air and 2 for exhaust air.

- Every activity must be documented.
- It involves those agents which can transmit through inhalation routes.
- It is applicable to clinical, diagnostic, research, and teaching and production systems.

57) What are precautions made in BSL2 as compared to BSL1?

- To work at Biosafety level 2, workers should be trained.
- Biosafety cabinet 1 is allowed at Biosafety level 2.
- Workers should be made immune before work.

58) Write 5 equipments which are used in BSL3.

- Vacuum lines must be protected with HEPA filter.
- Ducted air ventilation system must be used.
- Biosafety cabinet 2.
- HEPA filters on intake and exhaust air routes.
- Every action should be documented.

59) Describe two publications of bioethics.

Potter published his book Bioethics the survival of science. He described the bioethics as global movement.

Callahans published his book Bioethics as a discipline. He emphasized that bioethics should be taught as academic discipline.

Josef and Kennedy made an institution relating to human reproduction and bioethics which is now known as Kennedy Institute of Ethics.

60) What is relationship of nurses to their coworkers?

- Collaborative relationship
- To teach them

- To respect them

61) Explain case study of spleen removal.

A person is suffering from spleen cancer. He/she goes to physician for treatment. Physician suggests the only way to remove the spleen that is spleenectomy. Patient agrees for this treatment plan. When spleen was removed, the patient became healthy and went home. But physician used that spleen to find the gene responsible for cancer. He then requested for patenting that gene. After patenting he sold the patent rights to a company and earned a lot of incentives. Similarly the company is also using that gene for various researches and drugs.

Question arises here, whether it is ethical to use the spleen of patient without his informed consent.

There are two schools of thoughts about this. One is of the opinion that after organ removal from patient body, it no more belongs to the patient. For example in US this opinion is being practiced. On the other hand 2nd school of thought debates that it is the property of patients. Without informing or taking permission from patient it is unethical to use removed organs.

62) What are pharma crops? Give their use.

Pharmacrops or GM Crops are types of crops cultivated specially for the production of pharmaceutical drugs. They are not so common yet. Critics say that they may enter food chain and result in much harm to humans as well.

Final Term Questions:

63) Explain applications of Biosafety rule 2005.

1. Manufacture, import and storage of microorganisms:

a. Biosafety rules are applicable if we want to manufacture, import and store microorganism which are genetically modified.

2. Gene technological products for research

a. If we are dealing with genetically modified and genetically engineered organisms like pathogens we have to apply or follow rules of Biosafety 2005.

3. Field trials of GMOs

a. If we want to check the validity of GMOs in the fields for experimental purpose then we also have to follow Biosafety rule 2005.

4. Import, export, sale and purchase of GMOs

a. For import, export, sale or purchase genetically modified organism either animals, plants or pathogens we have to follow national biosecurity rules 2005.

64) Explain biosecurity plan.

Biosecurity plan is a written plan-prevent the introduction and spread of disease to farm. Daily operation procedures and disinfecting procedures are the part of the plan. Principal investigator has the responsibility to plan procedures and implement it. Workers must follow the plan, training for the lab workers, responsible official and of the campus security staff is required. Management services are also needed. Biosecurity officer must be contacted if biological agent is theft or lost, contact agencies if there is threat or spill. Other aspects such as risk assessment, physical protection, personal protection, pathogen accountability and emergency response should be considered.

65) How we immune the workers?

The risks of working with particular agents should be fully discussed with individual's researchers. The local availability, licensing state and utility of possible vaccines and therapeutic drugs like antibiotic treatment in case of exposure should be evaluated before work with such agents is started. If a particular vaccine or toxoid is locally licensed and available, it should be offered after a risk assessment of possible exposure and a clinical health assessment of the individual have been carried out. Facilities for specific clinical case management following accidental infections should be available.

66) Explain functions of technical advisory committee.

- Examine applications and recommend to NBC
- Review and control of safety measures
- Review research methodologies
- Monitor release of GMOs/products into environment

- Provide information to NBC about approved projects
- Supervise the implementation of terms and conditions

67) What are first aid equipments?

The minimum first aid equipment consists of a first aid box, protective clothing, and safety equipment for personnel and eye irrigation equipments.

1. First Aid Box:

- a. It keep the content dust and damp free.
- b. It should be kept in a prominent position and easily recognized.
- c. It is identified by a white cross and green background.
- d. It should contain instruction sheet for guidance, sterile adhesive dressings with different sizes, sterile eye pads, triangular bandages, sterile wound coverings, safety pins, unmediated wound dressings, authoritative first aid manual provided by Red Crescent.

2. Protective equipment:

- a. Mouth piece for mouth-mouth resuscitation
- b. Gloves and other barrier protection against blood exposure.
- c. Clean up kit for blood spills

3. Eye irrigation equipment should be available and staff trained in its correct use.

68) What are bioweapons? What are their threats? How we can mitigate them?

Bioweapons:

Infectious agents that are intentionally used against humans who cause harm to humans are known as bioweapons or bioterrorism. Bioweapons and bioterrorism are synonyms.

At national and international level plans are developed against bio-war defense. Bioterrorism affects public as well as scientific community because there is no vaccination and therapeutic agents against such weapons.

Biological threats:

1. Terrorist

a. These are non-state actors which potential use bioweapons to destabilize the governments.

2. Misuse of technology

a. When proper biological practices are not followed in labs, then infectious agents may spread in the environment causing harm.

3. Theft from lab

a. Biological agents can be theft from laboratories and used in unfair means.

4. Religious extremists

a. Some religious extremists can also use them against other states or groups.

5. Locally hired agents

a. Due to unemployment and poverty, international agencies can hire such people as agents which spread the infectious agents.

6. Frustrated cult

a. Similarly frustrated cults can also be used as agents against states.

Efforts to mitigate bioweapons:

1. Biological and toxin weapon convention (BTWC) act:

a. It was established in 1972. According to this there must be designation of focal point.

b. There must be central implementation authorities dealing with Biosafety and biosecurity guidelines.

c. There should be proper code of conduct for biologists and researchers.

d. Awareness among public about bioweapons, Biosafety and biosecurity is most important which can be through seminars, conferences and workshops, training and media.

2. Pakistan effort to mitigate biological threats:

a. Bioweapons are not the part of security matrix in Pakistan.

b. In 2005 national Biosafety rules have been made.

c. Administrative measures are also taken for risk assessment and management.

3. NGCL-2007:

a. According to this literature related to Biosafety should be in the syllabus.

4. PBSA – 2008

a. This deals with training of staff which seek biological certification. The modules of training are consisted of risk assessment, risk management and lab designing.

5. NIH-WHO:

a. It works in collaboration with WHO to control diseases and their spread.

b. Export control act 2004 for re-export and over export is also under NIH for transshipment of goods, technologies and equipment.

c. In case of guilty 14 years of imprisonment and 5 million rupees are fined.

69) What are the threats of biological weapons?

Biological threats:

1. Naturally present in environment

2. No major infrastructure needed

3. No manpower is needed

4. Easier and faster in action

5. Cost effective than poor man atomic bomb

6. Cover large area

7. Difficult to diagnose and treat

8. High mortality and mobility

9. Possibly contagious like small pox, plaque, viral hemorrhage fever

10. Create panic

11. Weapon of mass disruption

70) What risk assessment is made by GMOs?

· Inserted genes should have known properties whether they are coding for

Toxins/cytokines/hormones/allergens.

- Gene expression regulators/enhancers should be controlled.
- Sometime inserted genes can change into oncogenes sequences which cause cancer.
- Antibiotic resistance must also be assessed.

71) Explain different types of fire extinguishers.

Types of fire extinguishers:

1.. Water and foam:

a. Extinguish Class A fire by separating oxygen from triangle.

2.. Carbon dioxide:

a. Extinguish Class B and C by removing oxygen and heat from triangle.

3.. Dry chemical:

a. Extinguish Class A, B and C by interrupting chemical reaction

4.. Wet chemical:

a. Extinguish class K by removing heat

5.. Clean agents (Halogens)

a. Extinguish class A, B and C by interrupting chemical reaction.

6.. Water mist:

a. Extinguish class A by removing heat

72) What are functions of institutional Biosafety committee?

1. To assist the activities of NBC and technical advisory committee.
2. To assist researchers
3. Determine additional safeguards.
4. Evaluate qualification of researchers
5. Monitor work for Biosafety guidelines
6. Serve as a gateway for information between NBC and research teams
7. Update directory at every Biosafety level

8. Health of lab and field personnel
9. Contact with NBC and technical advisory committee for import and export
10. Prepare and implement emergency plans
11. Hold funds and to access projects that under which category it falls
12. Inspect and certify labs/plant gas house/ animals houses.

73) Explain general legislation with detail.

????

74) Explain Canada biosecurity legislation.

Canada biosecurity legislation deals with health of humans, animals and plants. It also deals with environmental safety issues. In this legislation two agencies are involved

Public health agency of Canada:

It deals with the health of human beings that no pathogen should enter into the Canadian environment that is lethal for human beings.

Canadian food inspection agency:

This agency gives the approval of genetically modified crops before entering into the market.

All these agencies are providing guidelines for human and animal pathogens and toxins to handle process or store them.

Guidelines:

If we are talking about Canadian biosecurity legislation that deals with the laboratory following guidelines must be

- Building must be fully equipped and must be designed in such a way that pathogens should not release from lab.
- All facilities must be there for processing, handling, storing or using pathogens or toxins.
- If we are handling with risk group 3 or 4 then pathogens must be handled in a Biosafety cabinet.
- Biosecurity legislation must update with the pathogens to improve the performance. It is most

important objective of Canadian biosecurity legislation.

75) What precautions are made in BSL2? How it differs from BSL1?

Precautionary measures made at BSL2:

1. Restricted movement of people

a. People are not allowed in BSL2. Children and old aged have weak immunity. Similarly animals are not allowed to enter in the lab.

2. Special bio-containment equipment

a. Biosafety cabinets are used in BSL2 for centrifuging, shaking, grinding, opening and even sonicating the samples. Aerosols formed can be filtered by two way air movements and HEPA filters in Biosafety cabinet.

3. Double doors

a. BSL2 labs have double doors for entry and exit.

4. Awareness about infectious agents

a. Workers working in the BSL2 must be made aware about infectious agents so that they may adapt standard microbiological measures while handling.

5. Immunization of workers

a. Staff must be immunized before working with infectious agents.

6. Biosafety manuals

a. Biosafety manuals with standard protocols and procedures must be present in the BSL2.

7. Proper sample collection

a. While collecting samples standard microbiological practices are performed. It is made sure that sample bags or containers must be leak proof.

8. Proper storage and transport of samples

a. Samples are stored at their relative temperatures. To transport them special and standard practices must be applied.

9. Decontamination after work

a. Decontamination of work area after completion of task must be decontaminated.

10. Proper spillage handling

a. In case of spillage, proper Biosafety measures should be followed including informing to Biosafety supervisors.

Difference between BSL1 and BSL2:

1. Major difference between Biosafety level 1 and Biosafety level 2 is that the workers which are working in Biosafety level 2 labs must have special training for handling the infectious material.

2. There should special supervisors for training of these workers which must be masters in the field.

76) Write Biosafety measurements in TB lab.

Biosafety measures to handle the sputum, blood or cultures. Biosafety measures can affect the

- code of practice,
- equipment,
- lab design and facilities,
- health surveillance,
- training and
- waste handling

In all these aspects we need special type of training.

77) Write elements of physical security.

Elements of physical security are of three types.

1. Obstacles

a. These are the barriers which physically stop attackers.

b. Fencing

c. Wall

d. Multiple locks

- e. Fire proof safes
- f. Water sprinkles
- 2. Notification system
 - a. These are heat detectors
 - i. Room temperature is 25C
 - b. Smoke detectors
 - c. Lighting
 - d. Alarms
 - e. Cameras
- 3. Recovery stage
 - a. Repairmen
 - b. Hiring additional security
 - c. Cameras

If security systems cannot be updated then more and more persons should be hired for security.

78) How risk can be reduced through risk management?

For risk reduction the term management is used which is a structured approach that occurs in steps. The purpose is to identify, assess as well as to control the risk. After identification we have to process in place to minimize the monitor the risk. Then we have to control the after effects of risk or the results that occur if risk happen or they should materialize.

Steps of risk management:

- First step is early consultation. If we have designed a lab then we have to identify our needs, our requirements and also we have to look at the cost.
- Second step is deferring irreversible decisions that to develop alternative ways to achieve our objectives or goals.
- Third one is the pilot studies. These are small studies. The purpose is to get more and more information about the risk so that it may be assessed or controlled.

- Fourth step is to make flexible designs which may be modified when future needs arises.
- Precautionary actions should be part of daily Biosafety and biosecurity systems. For example if the technician is transferring sample to lab and spillage occurs in the way due to leakage then we must take or follow precautionary measure or procedure of disinfection.
- Transferring risk to private sectors in the form of insurance is next step. It is better to use simple techniques instead of using complex techniques which led to less use of leading edge technology.
- Next step is to look at site investigation. What are foreseen grounded systems and what are their benefits.
- We have to review the project or program at every step. Cross examination may prove useful in case of valuable addition.
- It is better to stop the project when it is too risky for humans, animals, plants or environment.

79) Discuss malicious risk groups and types.

Malicious Risk Groups:

There are five malicious risk groups.

1. Non-pathogenic

a. These are the inherent hazardous agents or pathogens that causes no or insignificant consequences. These are not harmful for human beings.

2. Low malicious use risk (LMUR)

a. These are called low because when used maliciously they cause low consequences.

Most of the biological agents are included in these first two groups. For example mycobacterium leprae is an acid fast bacilli. They give strong acid fast reactions when stained with carbol fuchsin. The sides of these bacteria are parallel and ends are rounded and arrangements are usually palisade. This mycobacterium leprae cause a disease known as leprosy. Here organism grows very slowly and generation time is about 30 days. It is not highly virulent. We can treat

the patient by giving multi drug treatment. This microorganism is included in low malicious use risk (LMUR)

3. Moderate malicious use risk (MMUR)

a. These microorganism or pathogens or hazardous agents cannot be deployed as biological weapons. It causes low or moderate consequences. They cause low consequences on economy of the nations. Many current agents are evaluated as MMUR. For example *C.immitus* is a fungus. It causes a disease commonly known as desert fever. Desert fever apparently is asymptomatic means that no symptoms appear in the patient and it can be cured without any treatment. But some life threatening cases has been reported so it is included in Biosafety level 3 and it is considered to be MMUR.

4. High malicious use risk (HMUR)

a. It causes national or international consequences it means that these are the inherent hazardous agents that cause high consequences when used maliciously. They can be exploited as biological weapons. Such pathogens lead to high casualties and high economic impact. For example *bacillus anthracis* that causes a disease known as anthrax. This bacterium is a gram positive rod, spore former and in 2004 in Mary land in United States of America these spores were transferred from one person to another in the form of mailed letters. The mail letters contained the spores of *bacillus anthracis*.

5. Extreme malicious use risk (EMUR)

a. Sometime EMUR is considered as HMUR because it also causes high use risk when used maliciously. They are not found in the nature. Here high security measures are required. It should be eradicated from nature. For example if you have prepared genetically engineered agents it is included in EMUR. It should be eradicated from the lab as well as from experimental procedures. For example *Varioloa major* virus which causes small pox. Soviet Union has used small pox as a bioweapons. After that it was decided that small pox should be eradicated from the world by vaccination. *Variola* virus is highly virulent, contagious and stable in aerosol or

droplets. Scientists have tried to genetically modify this virus from which it was found that genetically modified variola is more virulent so they must be eradicated from the world by simple use of vaccines.

80) What are functions of physical security?

Protection of people/hardware/programs:

It is the first component of biosecurity. It not only gives protection to the people working in lab but also protect the hardware and programs with which laboratory or organization is dealing.

Data or Networks:

If the any kind of data is available in computer, what measures are must be taken to secure it.

Similarly the procedures followed in the lab must also be secured.

Physical events:

Physical circumstances or events can occur in the form of terrorism like theft of pathogens or toxins.

Natural disaster in the form of fire or explosive chemicals is to be dealt. Physical security not only deals with protection of lab workers or hardware but it also protects data, network and software programs. So to deal with bioterrorism or natural disasters we use physical security component of biosecurity.

81) What are surveillance elements?

Surveillance or Notification system

a. These are heat detectors

i. Room temperature is 25C

b. Smoke detectors

c. Lighting

d. Alarms

e. Cameras

82) Name lab safety things.

- Biological safety cabinets
- Negative pressure flexible film isolators
- Micro-incinerators
- Vacuum line protection
- Others
- o Pipetting aids
- o Spatter shields
- o Autoclave
- o Screw capped bottles
- o Disposable loops

83) Discuss transport security elements.

Transport security is consisted of two elements.

1. Internal transport

- a. Transport or movement of biological material within the country.
- b. Movement from or to restricted area but within facility.
- c. If we are moving biological material within the institution or within the city even we will refer a reliable lab worker.
- d. If we are moving biological material from one city to another we divide the area into three groups
 - i. Shipping area (ID card and special documentation of pathogens or agents)
 - ii. Receiving area (Receiver name, ID card, Code answer of Code question)
 - iii. Disposal area (Incinerators)

2. External transport

- a. Transport of movement of biological material to another country.
- b. Transport or movement from one facility to another.

- c. This involves commercial carriers
- d. More than 30 commercial carriers are available
- e. Most commonly used are IATA and WHO
- f. We will transport frozen samples
- g. To maintain freezing we have to use gas
- h. It is expensive method as compared to internal
- i. For external transport we divide the material into two groups
 - i. Category A
 - ii. Category B (Infectious material like cultures, isolated DNA, RNA or Nucleic Acids)
- j. Such infectious material requires triple packaging system with primary, secondary and tertiary layers. Primary consist of specimen who protect it from leakage, second layer protects the primary layer which is water proof and third layer protects the secondary layer.

84) Write activities of Biosafety officer.

Activities of Biosafety officer:

Activities of Biosafety officer include Biosafety consultation, biosecurity consultation and technical compliance consultation.

Among other activities of Biosafety officer internal audits are also part. Audit of Biosafety make sure that every worker is conscious about following the Biosafety protocols. Internal Biosafety audits are very necessary regarding technical methods, procedures, protocols, infectious or biological agents, materials and equipments. If any of the equipment is sent out for repair it should be decontaminated prior to departure.

If the person or worker has violated the Biosafety protocols then it is the duty of Biosafety officer to discuss with the worker who has violated. Biosafety training should be done after every three months or six months to ensure the continuity of Biosafety education. In case of any accident in the lab, complete investigation should be done about accident and it is duty of Biosafety officer to prepare the report

within 10 days.

In case of any spillage, it is the duty of Biosafety officer to decontaminate the floor properly either it was toxin or blood samples or chemicals. Proper waste management is also the duty of Biosafety officer. For example use of incinerators for larger disposal and sterilization for smaller cultures may be done in autoclaves. Decontamination or sterilization of apparatus can be done in an oven or in an autoclave. If we want to sterilize in oven then we have to stick to the temperature that is 80C.

Awareness of community attitudes is very necessary regarding the health and environmental issues.

We also have to transport infectious material for which Biosafety officer must be conscious that whether we are following the IATA guidelines or WHO guidelines. Research proposals comes to the Biosafety officers they can review and send their own suggestions.

The last duty of Biosafety officer is to develop an institution of the system to deal with emergencies. If institution is making plans about dealing with emergencies, then Biosafety officer should be part of discussion or committee.

85) Explain BSL4

It is the highest containment level and it includes those agents that are dangerous exotic because they have the ability to appear in the environment and contaminate the environment in the form of aerosols. Aerosols can easily be inhaled by workers who are working in the lab and they can be easily infected by the infectious agents. Dangerous exotic agents are those which can cause lethal disease in humans and no special vaccines are available, no treatment or even no precautionary measurements. Here we have to take special precautionary measures for the workers who are working in the lab in order to prevent the formation of aerosols as much as possible. There is also unknown risk of transmission due to which we have divided Biosafety level 4 into two types.

1. Cabinet laboratory:

a. We can manipulate cultures, infectious material in a Biosafety cabinet. Biosafety cabinets which are certified and are used according to manufacturer's detail have double doors. Materials which are infectious first pass through autoclave sterilization

and then it is then disposed off from Biosafety cabinet. In autoclave sterilization occur in 121C for 15 to 20 Minutes with a pressure of 15psi. Biosafety cabinet must be designed in such a way that if infectious material cannot be sterilized by autoclaving then it must be removed carefully from the Biosafety cabinet. Never put sharp things in the Biosafety cabinet. Apart from HEPA filters which are used for filtration of intake air and exhaust air but gas and vapor decontamination unit is also there. Workers using Biosafety cabinet must wear protective equipments like gloves, eyes sheets, shoe covers and all other safety equipments which are required by Biosafety level 4. It is the responsibility of lab manger to repair and maintain the Biosafety cabinet. Biosafety cabinet maintenance should be done daily.

2. Suite laboratory

a. In this type of Biosafety level 4, a worker wear a positive pressure air protective suit but all the microbiological procedures are performed in the Biosafety cabinet.

Decontamination of the all the equipments and suits is done during the lab operations.

86) What are symptoms of SARS?

In general SARS patients present with fever in which temperature great than 100.4F, malise and myalgias quickly followed by respiratory symptoms including shortness of breath and cough. 10 to 20% of patients may have diarrhea. Review of probable cases indicates that the shortness of breath sometimes rapidly progress to respiratory failure requiring ventilation. The case fatality rate is about 11%.

87) Write a note on influenza virus.

Influenza is an acute viral disease of the respiratory tract. The most common clinical manifestations are fever, headache, malaise, sore throat and cough. GI tract manifestations (nausea, vomiting and diarrhea) are rare but may accompany the respiratory phase in children. The two most important features of influenza are the epidemic nature of illness and the mortality that arises from pulmonary complications of the disease. The influenza viruses are enveloped RNA viruses belonging to the

Orthomyxoviridae. There are three serotypes of influenza viruses, A, B and C. Influenza A is further classified into subtypes by the surface glycoproteins that possess either hemagglutinin (H) or neuraminidase (N) activity. Emergence of completely new subtypes (antigenic shift) occurs at irregular intervals with Type A viruses. New subtypes are responsible for pandemics and can result from reassortment of human and avian influenza virus genes. Antigenic changes within a type or subtype (antigenic drift) of A and B viruses are ongoing processes that are responsible for frequent epidemics and regional outbreaks and make the annual reformulation of influenza vaccine necessary. Influenza viral infections, with different antigenic subtypes, occur naturally in swine, horses, mink, and seals and in many domestic and wild avian species. Interspecies transmission and reassortment of influenza A viruses have been reported to occur among humans and wild and domestic fowl.

88) What was the name of first test tube baby? When he was born?

Louise Brown in 1978

89) What are duties of fire wardens?

To assist in implementing and improving effective emergency procedures in your workplace;

- To help prevent emergencies by monitoring the adequacy of the fire risk control

Measures;

- To raise awareness with other staff about the fire hazards that exist in your workplace;
- To instruct workers in how to respond in an emergency;
- To lead the fire drills and real evacuation procedures – they must be familiar with all

Escape routes and exits from their designated area;

- To ensure all workers are accounted for during an evacuation; and
- To assist all people in the workplace should an emergency occur, including assisting people with special needs, e.g. helping someone in a wheelchair to evacuate.

90) Explain personal security.

Personnel security is a system of policies and procedures which seek to manage the risk of staff

(permanent, temporary or contract staff) exploiting, or intending to exploit, their legitimate access to an organization's assets or premises for unauthorized purposes. Although many organizations regard personnel security as an issue resolved during the recruitment process, it is a discipline that needs to be maintained throughout a member of staff's time in employment. This includes robust pre-employment screening, effective line management, employee welfare, clear lines of communication, and a strong security culture. It should also include a formal process for managing staff leaving the business. When applied consistently, personnel security measures not only reduce operational vulnerabilities, they can also help build a hugely beneficial security culture at every level of an organization. Robust personnel security helps organizations to employ reliable people; minimize the chances of staff becoming unreliable once they have been employed; and detect suspicious behavior and resolve security concerns once they emerge.

91) Write five characteristics of NDV (Newcastle Disease Virus)

ND is one of the most serious infectious diseases of poultry worldwide. It is primarily a respiratory disease, but nervous and enteric forms occur. All bird species are probably susceptible to infection with ND virus (NDV). The severity of the disease caused by any given NDV strain can vary from an unapparent infection to 100% mortality. The chicken is the most susceptible species

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Q1: Facilities and Apparatus used in BSL-1 2

Safety Equipment

We require following equipment which are used in all general labs. For example

§ If we are performing very simple biological experiment we have to first wear the gloves because we are handling infectious agents. If the gloves are disposable then dispose them off after performing the procedure and never wear the disposable gloves again.

§ It is better to use lab coat, apron or any type of protective clothing while working in lab.

White lab coat should be used. Their cuffs should have elasticity and belt at the back, having full sleeves.

§ Similarly if we are dealing with any chemical that has fumes and it is very toxic, it can cause skin and eye irritation, then it is necessary to use protective eye wear while opening the chemical in order to avoid any type of eye irritation.

Laboratory facilities:

- ✓ Lab doors must be kept control and there must be proper doors for access control.
- ✓ There should be deep sink for hand washing.
- ✓ Bench top should be made with a material that must be resistant to chemicals to organic solvents and inorganic solvent.
- ✓ Similarly chairs are made up of such a material that we can wash them easily even disinfect them.
- ✓ Lab windows that open external air must be fitted with screens.

Q2: Surveillance and Notification system elements 3

Surveillance and notification system:

These are the heat detectors heat detectors can give us alarms there is high temperatures for samples as well as equipments. Then smoke detectors detect the smoke it protect the building from fire. Light is proper in lab especially in the bench work. Then emergency alarms should be there, there should be cameras to control the access of unauthorized people to the lab.

Q3: Define Biosecurity 3

A set of preventive measures designed to reduce the risk of transmission of infectious agents

- Security against the inappropriate use of potentially dangerous biological agents

Q4: What are the activities of human which changed fresh water into sewage water? 2

Human activities change clean water into waste water. It can be domestic activity; it can be Agricultural, commercial, industrial, storm water, or it can be due the runoff water.

Q5: Potential threats to bioweapons 3

Biological threats:

1. Terrorist

a. These are non-state actors which potential use bioweapons to destabilize the governments.

2. Misuse of technology

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a. When proper biological practices are not followed in labs, then infectious agents may spread in the environment causing harm.

3. Theft from lab

a. Biological agents can be theft from laboratories and used in unfair means.

4. Religious extremists

a. Some religious extremists can also use them against other states or groups.

5. Locally hired agents

a. Due to unemployment and poverty, international agencies can hire such people as agents which spread the infectious agents.

6. Frustrated cult

a. Similarly frustrated cults can also be used as agents against states.

Q6: How the Radiation effect can be minimized

Protection: In order to avoid the risk of ionizing radiation we have to take certain protective measures.

We can reduce the time of exposure to the radiation. Similarly we can maximize or increase the distance from radiation source. If you are working in the radioactive Isotopes it is better to use the spatter Shield, i.e. plastic shield between the source and the worker. And if your technique is in such a way that instead of using ionization radiations, or radioactive isotopes, radio nucleotides, use the non radioactive metrics techniques or material which have more specificity

and sensitivity.

Q7: What are the malicious risk groups, what are their abbreviations explain in detail. 10

Malicious Risk Groups:

There are five malicious risk groups.

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Q8: Applications of National Biosafety Rules

Applications with reference to national biosafety rules 2005

National biosafety rule 2005 are applicable to manufacturing of genetically modified product for example national biosafety rule 2005 has divided into different committees; one is the institutional biosafety committee, 2nd one national biosafety committee, 3rd one is technical advisory committee.

The applications of national biosafety rules are divided among these committees they hold the

funds they give projects to institution. And if they want to manufacture a product from the institution by using rDNA technology then here national biosafety rules are applicable.

If we are using a technique of a genetic engineering and we want to import some chemical or some biological material from other countries then again national biosafety rules are applicable here. Storage of microorganism to avoid hazards of bioweapons national biosafety rule 2005 is applicable. If you are using a product of gene technology for research purposes or your project deals with genetically modified organisms such as GM plants, GM animals and you are performing their field trial then national biosafety rule 2005 is applicable. If you prepared a product from rDNA and you want to export or import it again you have to follow the rules. Even for the sales and purchase of GMO we have to follow the national biosafety rule 2005.

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Q9: Japan Biosecurity Legislation

Japan ministry of Health, Labor and Welfare is involved to prepare these legislations.

Actually this ministry deals with the two pillars of Biosecurity one is Surveillance of Infection and Infectious Diseases, that how infectious agent should be processed, handled and stored.

Now the 2nd pillar of Biosecurity which is designed by the Japan ministry of health labor and welfare one is the Regulation of Pathogens Handling. That there must be special establishment if the procedures and there must be proper regulations and implementation of those procedures as well.

Duties:

Japan has created different ways they screen food, even human beings, vectors at point of entry into the Japan. So food vector or humans being screened whenever they enter to the Japan. The Japan ministry of agriculture forest and fisheries deals with health issues of animals and plants as well. Biosecurity legislation without proper control of bioweapons is incomplete. So Japan Biosecurity legislations, like the legislation of US and Canada, it also deals with bioweapons prohibition laws. Due to these laws bringing bioweapons in the state of Japan is strictly

prohibited.

So if we want to deal with the biopathogens that are lethal for the human being for research purpose then we have to take the special permission from the Japan ministry so that they can enter into the Japan state. Otherwise bioweapons which are used in the terrorism they are totally prohibited for entry into the Japan.

Q10: Objectives of Lab Biosecurity

Objectives:

Those laboratories which are dealing with the risk group 3 or 4 they mainly design Biosecurity objectives. The 1st lab Biosecurity objective is to design the safety agenda.

This support biosafety agenda to prevent disease. It is not only to safe environment but also prevents the spread of pathogens to the humans being animals and plants that ensure containment of infectious material. For this reason responsibilities are increased on the lab investigators Biosecurity Officer they must look at Biosecurity system of lab on routine basis. For the there must be spaces between the bench work, gloves should be available on each bench, each workers should wear lab coats, and other sanitation system should be improved.

The 2nd main objective of the lab Biosecurity is to the Maintain citizen confidence on bioscience research community. Now you have designed the product in the lab that is very useful if we market it, so it should be your experiment your product composition should be transparent to the investors in the industry. So this not only protects the valuable research but it is also commercially very important for you and organization and also for the economy of any country.

Then the last but very important objective of the lab Biosecurity is to protect the risk of crime any sort of crime and even of bioterrorism, there should no release of bioweapons in environment. For that there must be proper transport guidelines available by AITA and WHO. So if the nation has not its own transport guidelines then they can follow the International guidelines for the transport of infectious material.

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Q11: Biosecurity Plan 3

Biosecurity plan:

It is documentation or a written plan prevents introduction of pathogens into the nation that can affect any population or subpopulations and can even affect the organizations. It not only deals production of pathogen in to the environment or organizations also prevent spread of diseases. In this case we have to look at the daily operations procedures, these daily operation procedures also deals with disinfection method which is an important part of Biosecurity plane. So Biosecurity plane start with disinfection which is an important step of any procedure.

Q12: Potential Threats to Bioweapons 3

Potential threats:

- naturally present in the environment
- no major infrastructure
- no manpower
- easier and faster
- cost-effective than poor's man atomic bomb
- cover large area
- difficult to diagnose and treat
- high mortality and mobility
- possibly contagious small pox, plaque, viral hemorrhagic fever
- create panic
- weapon of mass disruption
- detection devices, equipment for surveillance expensive
- not present in many countries

Q13: Equipments in first Aid 5

Contain:

First aid must contain aid box, protective clothing (contain gloves, eye shield, face mask) safety equipment and eye irrigation. Safety equipment can be in the form of rope for fire hazards, mouth to mouth pipe for resuscitation then eye irrigations system must be properly placed in the first aid box.

First aid Box:

It is a special box made up of such a material that keep the contents dust and damp free. It has many symbols in different countries white cross with green box in some countries black and

some countries red and white. So symbol is varying depending upon the country in which we are using the first aid box. If we are dealing with laboratories any sort of accident occurs then first aid box placed in prominent position and easily recognizable because of their symbol. Must have first aid manuals. It should contain simple things like cotton, paper towel in order to clean spillage, the patient bandages of different sizes like triangular and Sterilize dressing and safety pins safety pins also in box for dressing.

Protective equipments:

It includes Mouth piece for mouth to mouth resuscitation then gloves are Important in order to protect hands from carrying infectious material. Clean up kit for blood spills or any container spillage, there must be a disinfectant.

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Q14: Functions of Technical Advisory Committee 5

Function of Technical Advisory Committee:

Is to examine applications researchers about the project, proposals and recommend it to national biosafety committee for their approval.

They review the safety measures as well, and they also control the safety measure after the approval of the project.

They Review the research methodology as well.

They also monitor commercial release of GMO and products into the environment.

They look at safety regarding humans health and environment as well.

They provide information regarding the national biosafety committee about approved projects up to which level project have been completed, the funds has been utilized, whether the article has published in international journal or not.

They also supervise the implementations of terms and conditions. Because whenever an application is approved or a proposal project is approved it is always following the terms and conditions.

Q15: Define BSL-2 2

Moderate hazards to the personal and environment. Microbial practices same to BSL1. Lab door must be closed. Smoking, drinking is not allowed in lab. Mouth pipetting is prohibited. Infectious material should be disposed properly.

Q16: Which are additional facilities in BSL4 as compared to BSL3? 2

We divided BSL-4 in two types

1) A cabinet laboratory: Manipulation of agent in BSC. We can manipulate culture infectious material in a biosafety cabinet. Now biosafety cabinet that is satisfied and used according to the manufacturer details they have double doors. Material that contains infectious agent first passes through autoclave for sterilization then disposed of biosafety cabinet. Now in autoclave sterilization occurs at 121 degrees Celsius for 15 to 20 minutes at pressure 15 pounds. Biosafety cabinet must be designed if infectious material not sterilized in autoclave it must be removed carefully from biosafety cabinet. Never put sharp things in BSC. It is better to avoid such material. Now apart from HEPA filters which are used in filtration that intake air and exhaust air gas and vapor decontamination unit in BSC. Now the worker who are using biosafety cabinet must wear glass eye shields then shoes cover once they enter into the lab they have followed all those personal equipments which are required for the lab BSL4. It is responsibility of lab manager to look at repair and maintenance of BSC.

2ND TYPE: A suit laboratory:

Where personnel must wear positive pressure air protective suit. BSC used for all microbiological procedures and decontamination of gloves and apparatus during lab operations.

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Q17: How risk can be reduced through risk management, describe the logical process 5

Risk Management:

Now for risk reduction the term risk management is used. This is structured approach that occurs in steps to identify assess and control the risk. After identification of the risk we have to

process them in a place where we can minimize and control the risk. We have to control the after effects of the risk and they should materialize.

Steps:

So the risk management that occurs in steps; the 1st step is early concentration.

For example if we have designed a lab then we have to identify our needs our requirements and also we have to look at cost.

The 2nd step is the Deferring irreversible decision we need some more time develop alternative ways in order to achieve our goals. So this is again a better approach to take more time to develop different techniques to achieve the objectives.

The 3rd one is the Pilot studies: these are small studies; the purpose of these small studies is to get more and more information about risk.

Then the 4th one is that we have to design such a strategies in risk assessment methodology that are flexible. So if the method is Design in such a way that we can easily modify them according to the future challenges.

5th one is the Precautionary actions should be the part of the daily biosafety and biosecurity system. E.g. if a technician transferring samples from collection center to the lab, and blood spillage occur in the way, so to manage this risk we need to take precautionary measures that is procedure if disinfection.

Then the next 6th step is Transferring of Risk to the private sector; in the form of insurance we can transfer the risk to the insurance companies. Now it is better to use simple technique instead of using complex techniques. So less use of leading edge technology.

Then in 7th step we have to look at the site investigation.

Then in 8th step if the management system is in the form of programs or in the form of projects, then before implementing them we have to review the project at every step. So that the same person can review it who is related to this discipline. Besides this it can be reviewed be the person of other disciplines due the reason that they can give their point of view as public.

Then in the last 9th step it is better to stop the project when it is too risky for human being for animals plants as well as environment.

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Q18: What are Bioweapons, What are their biological threats, and also explain the efforts to mitigate them. 10

Bioweapons:

Infectious agents that are intentionally used against humans who cause harm to humans are known as bioweapons or bioterrorism. Bioweapons and bioterrorism are synonyms.

At national and international level plans are developed against bio-war defense. Bioterrorism affects public as well as scientific community because there is no vaccination and therapeutic agents against such weapons.

Biological threats:

1. Terrorist

a. These are non-state actors which potential use bioweapons to destabilize the governments.

2. Misuse of technology

a. When proper biological practices are not followed in labs, then infectious agents may spread in the environment causing harm.

3. Theft from lab

a. Biological agents can be theft from laboratories and used in unfair means.

4. Religious extremists

a. Some religious extremists can also use them against other states or groups.

5. Locally hired agents

a. Due to unemployment and poverty, international agencies can hire such people as agents which spread the infectious agents.

6. Frustrated cult

a. Similarly frustrated cults can also be used as agents against states.

Efforts to mitigate bioweapons:

1. Biological and toxin weapon convention (BTWC) act:

- a. It was established in 1972. According to this there must be designation of focal point.
- b. There must be central implementation authorities dealing with Biosafety and biosecurity guidelines.
- c. There should be proper code of conduct for biologists and researchers.
- d. Awareness among public about bioweapons, Biosafety and biosecurity is most important which can be through seminars, conferences and workshops, training and media.

2. Pakistan effort to mitigate biological threats:

- a. Bioweapons are not the part of security matrix in Pakistan.
- b. In 2005 national Biosafety rules have been made.
- c. Administrative measures are also taken for risk assessment and management.

3. NGCL-2007:

- a. According to this literature related to Biosafety should be in the syllabus.

4. PBSA – 2008

- a. This deals with training of staff which seek biological certification. The modules of training are consisted of risk assessment, risk management and lab designing.

5. NIH-WHO:

- a. It works in collaboration with WHO to control diseases and their spread.

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- b. Export control act 2004 for re-export and over export is also under NIH for transshipment of goods, technologies and equipment.
- c. In case of guilty 14 years of imprisonment and 5 million rupees are fined.

Q19: Duties of Biosafety Committee 3

Then the duty of the biosafety committee is the risk assessment or formulation of proper biosafety policies, risk assessment either in the form of forms or it can be depending on the risk

group that whether the institution is dealing with risk group 1 2 3 or 4.

If Biosafety committees dealing with sensitive issues then we can discuss it with the experts of biosafety.

So biosafety is not a ring of a single member but it is a group of different people coming from the different disciplines. Either it can be biosafety officers, scientists, medical officers, Veterinary Doctors, or technical staff, even the management staff can be a member of biosafety committee. So that all the problems related to biosafety can be discussed. If fire erupts anywhere then fire extinguisher supervisor can be the member of biosafety committee.

Q20: Risk of Ionization radiations 5

Risk:

Now why it is so much dangerous to use ionizing radiation? Why in this case biosafety is required? This is because of 2 effects

Somatic Effect occurs when the radiation caused or induced cancer it can mutate protooncogene into oncogenes and workers can suffer from leukemia, Blood cancer lung cancer or even bone cancer.

Hereditary affect of radiation exposure to the gonads can occur. It is very much dangerous again.

Protection: In order to avoid the risk of ionizing radiation we have to take certain protective measures.

Q21: What comprises of containment Zone? 2

Different biosafety levels required for the containment zone. Whenever we talk about the containment zone, it can be only a chemical fume hood. That is required for isolation identification and characterization of microorganisms such as bacteria.

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Q22: Activities of Biosafety Officer 5

Activities

Biosafety officer perform different activities, it can be biosafety or biosecurity consultation, it can be technical compliance consultation.

Biosafety audits: Other activities include internal biosafety audits. So that every worker should be conscious that he or she following the biosafety protocols. Internal biosafety audits are very necessary regarding technical methods, procedures protocols, procedures it can be infectious or the biological agents, material source or even audit is acquired for the equipments.

Activities

If the person violate the biosafety protocols that it is duty of biosafety officer to discuss with that person whether he she not dealing with techniques properly. This is not only the work of the biosafety officer but it should also be done by the laboratory supervisor. Biosafety training should be done after 3 three months or after 6 months so that continue education in biosafety. If there is any accident in the lab happened then a complete investigation should be done. And it is the duty of the safety officer to prepare the report within 10 days.

Decontamination of spillage e.g. spillage of sample occur such as blood, spillage due to toxic chemicals it is duty of officer that floor must be decontaminated properly. Proper waste management should be there in the laboratory, it can be through incinerators but if it occur at shorter scale e.g. if we sterilize the culture then it can be done in an autoclave.

Then decontamination of then apparatus, sterilize it in an oven or place it in autoclave. If we want to sterilize the apparatus in an oven then we have to stick to the temperature i.e. 80 degrees Celsius.

Then awareness of community attitudes is very necessary regarding health environmental issues. We also have to deal with transport of infectious material, biosafety officer must be conscious that whether we are following the AITA guidelines or WHO guidelines properly or not.

Then research proposals come to the biosafety officers, they can review and they can send their own suggestions. The last activity that has to be performed by the biosafety officer is that is the

Institution of a system to deal with emergencies. So whenever the institution is making emergency plans then the committee must comprise of biosafety officer as well.

Q23: Equipment used in BSL-3 5

Equipment: now here special equipments are required for example apart from personal equipment glasses eye face mask shoe cover we also require special vacuum line must be protected with HEPA filters. We need ducted air ventilation system. In biosafety cabinet HEPA filters used same in ducted air ventilation system. Biosafety cabinet actually a chamber that has three filters 1 HEPA filter intake air and 2 HEPA filter for exit air in order to disperse away the contaminated air from occupied area. BSL 3 must be fully documented facility design that include decontamination of larger pieces of equipments any sort of operation any parameter regarding procedure must be documented prior to performing the procedure.

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Q24: Duties of Japan Biosecurity Agency 5

Japan ministry of Health, Labor and Welfare is involved to prepare these legislations. Actually this ministry deals with the two pillars of Biosecurity one is Surveillance of Infection and Infectious Diseases, that how infectious agent should be processed, handled and stored. Now the 2nd pillar of Biosecurity which is designed by the Japan ministry of health labor and welfare one is the Regulation of Pathogens Handling. That there must be special establishment if the procedures and there must be proper regulations and implementation of those procedures as well.

Duties:

Japan has created different ways they screen food, even human beings, vectors at point of entry into the Japan. So food vector or humans being screened whenever they enter to the Japan. The Japan ministry of agriculture forest and fisheries deals with health issues of animals and plants as well. Biosecurity legislation without proper control of bioweapons is incomplete. So Japan Biosecurity legislations, like the legislation of US and Canada, it also deals with bioweapons

prohibition laws. Due to these laws bringing bioweapons in the state of Japan is strictly prohibited.

So if we want to deal with the biopathogens that are lethal for the human being for research purpose then we have to take the special permission from the Japan ministry so that they can enter into the Japan state. Otherwise bioweapons which are used in the terrorism they are totally prohibited for entry into the Japan.

Q25: Define Cybernetics 2

Answer:- The study of interactions between man, machine and animals is called cybernetics. It is a Greek word “governance”. It is exploring regulatory system, their structures and functions.

Q26:Physical Security elements? 3

Answer: It not only deals with protection of people but Also deals with hardware network programmers as well as protection of data .Now physical elements are of three types these are;

- obstacles
- notification and
- recovery stage.